



Top 77 CAT Number Series Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

The number of common terms in the two sequences 17, 21, 25,..., 417 and 16, 21, 26,..., 466 is

- A 78
- B 19
- C 20
- D 77
- E 22



 VIDEO SOLUTION

Question 2

The sum of 3rd and 15th elements of an arithmetic progression is equal to the sum of 6th, 11th and 13th elements of the same progression. Then which element of the series should necessarily be equal to zero?

- A 1st
- B 9th
- C 12th
- D None of the above

The sum of 3rd and 15th elements of an arithmetic progression is equal to the sum of 6th, 11th and 13th elements of the same progression. Then which element of the series should necessarily be equal to zero?

A) 1st
B) 9th
C) 12th
D) None of the above

a, d
 $T_n = a + (n-1)d$
 $S_n = \frac{n}{2} \{2a + (n-1)d\}$

$a + 2d + a + 14d$



VIDEO SOLUTION

Question 3

The 288th term of the series a,b,b,c,c,c,d,d,d,d,e,e,e,e,f,f,f,f,f... is

- A u
- B v
- C w
- D x

The 288th term of the series a,b,b,c,c,c,d,d,d,d,e,e,e,e,f,f,f,f,f... is

A) u
B) v
C) w
D) x

$a \rightarrow x^1$
 $n(x+1) \geq 288$

A-1	1
B-2	3
C-3	6
D-4	10
E-5	15



VIDEO SOLUTION

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Question 4

There are 8436 steel balls, each with a radius of 1 centimeter, stacked in a pile, with 1 ball on top, 3 balls in the second layer, 6 in the third layer, 10 in the fourth, and so on. The number of horizontal layers in the pile is

- A 34
- B 38
- C 36

D 32

There are 8436 steel balls, each with a radius of 1 centimeter, stacked in a pile, with 1 ball on top, 3 balls in the second layer, 6 in the third layer, 10 in the fourth, and so on. The number of horizontal layers in the pile is

A) 34
B) 38
C) 36
D) 32

VIDEO SOLUTION

Question 5

If the product of n positive real numbers is unity, then their sum is necessarily

- A a multiple of n
- B equal to $n + 1/n$
- C never less than n
- D a positive integer

If the product of n positive real numbers is unity, then their sum is necessarily

A) a multiple of n
B) equal to $n + 1/n$
C) never less than n
D) a positive integer

VIDEO SOLUTION

Question 6

In a certain examination paper, there are n questions. For $j = 1, 2, \dots, n$, there are 2^{n-j} students who answered j or more questions wrongly. If the total number of wrong answers is 4095, then the value of n is

- A 12
- B 11
- C 10
- D 9

VIEW SOLUTION



Question 7

If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

- A 0
- B -1
- C 1
- D Not unique

A video player interface showing a handwritten solution for Question 7. The text on the screen reads: 'If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?'. Below this, the calculations are shown: $\frac{11}{2} \{2a + 10d\} = \frac{19}{2} \{2a + 18d\}$, which simplifies to $11(a + 5d) = 19(a + 9d)$. Further down, the formulas for the nth term and the sum of the first n terms are given: $T_n = a + (n-1)d$ and $S_n = \frac{n}{2} \{2a + (n-1)d\}$. The cracku logo is visible in the top right corner of the video frame.

VIDEO SOLUTION

Question 8

On January 1, 2004 two new societies S1 and S2 are formed, each n numbers. On the first day of each subsequent month, S1 adds b members while S2 multiplies its current numbers by a constant factor r. Both the societies have the same number of members on July 2, 2004. If $b = 10.5n$, what is the value of r?

- A 2.0
- B 1.9
- C 1.8
- D 1.7

VIDEO SOLUTION

Question 9

Consider the sequence of numbers $a_1, a_2, a_3, \dots$ to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?

- A -100.33

- B -30.00
- C 62.33
- D 119.33

Consider the sequence of numbers $a_1, a_2, a_3, \dots$ to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?

A) -100.33
B) -30.00
C) 62.33
D) 119.33

Handwritten solution showing the sequence terms and their sum:

$$a_3 = a_2 - a_1$$

$$a_4 = a_3 - a_2 = -a_1$$

$$a_5 = a_4 - a_3 = -a_1 - (-a_1) = 0$$

$$a_6 = a_5 - a_4 = 0 - (-a_1) = a_1$$

$$a_7 = a_6 - a_5 = a_1 - 0 = a_1$$

VIDEO SOLUTION

CAT Previous Papers
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Question 10

Consider a sequence where the n^{th} term, $t_n = n/(n+2)$, $n = 1, 2, \dots$. The value of $t_3 * t_4 * t_5 * \dots * t_{53}$ equals.

- A 2/495
- B 2/477
- C 12/55
- D 1/1485
- E 1/2970

Consider a sequence where the n^{th} term, $t_n = n/(n+2)$, $n = 1, 2, \dots$. The value of $t_3 * t_4 * t_5 * \dots * t_{53}$ equals.

A) 2/495
B) 2/477
C) 12/55
D) 1/1485
E) 1/2970

Handwritten solution showing the sequence terms and their product:

$$t_3 = \frac{3}{5}$$

$$t_4 = \frac{4}{6}$$

$$t_5 = \frac{5}{7}$$

$$t_6 = \frac{6}{8}$$

$$\vdots$$

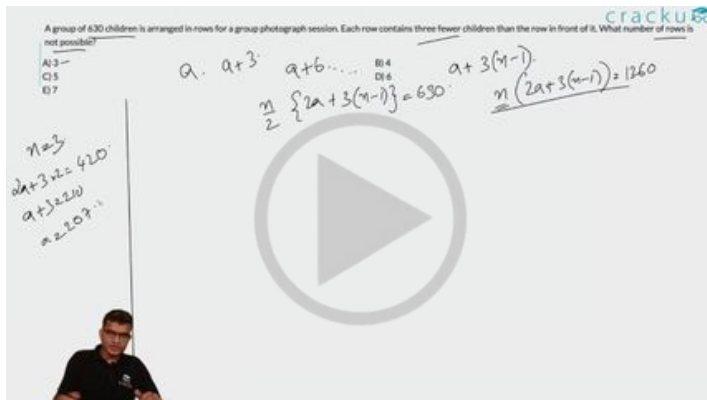
$$t_{53} = \frac{53}{55}$$

VIDEO SOLUTION

Question 11

A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. What number of rows is not possible?

- A 3
- B 4
- C 5
- D 6
- E 7



VIDEO SOLUTION

Question 12

Let $x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots}}}}$ to infinity. Then x equals

- A 3
- B $(\sqrt{13} - 1)/2$
- C $(\sqrt{13} + 1)/2$
- D $\sqrt{13}$

VIEW SOLUTION



CAT Syllabus PDF



Question 13

Two men X and Y started working for a certain company at similar jobs on January 1, 1950. X asked for an initial monthly salary of Rs. 300 with an annual increment of Rs. 30. Y asked for an initial monthly salary of Rs. 200 with a rise of Rs. 15 every 6 months. Assume that the arrangements remained unaltered till December 31, 1959. Salary is paid on the last day of the month. What is the total amount paid to them as salary during the period?

- A Rs. 93,300

- B Rs. 93,200
- C Rs. 93,100
- D None of these

VIDEO SOLUTION

Question 14

For a Fibonacci sequence, from the third term onwards, each term in the sequence is the sum of the previous two terms in that sequence. If the difference in squares of 7th and 6th terms of this sequence is 517, what is the 10th term of this sequence?

It is known that all the terms of the Fibonacci sequence are natural numbers

- A 147
- B 76
- C 123
- D Cannot be determined

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For a Fibonacci sequence, from the third term onwards, each term in the sequence is the sum of the previous two terms in that sequence. If the difference in squares of 7th and 6th terms of this sequence is 517, what is the 10th term of this sequence?

It is known that all the terms of the Fibonacci sequence are natural numbers

A) 147
B) 76
C) 123
D) Cannot be determined

VIDEO SOLUTION

Question 15

If $a_1 = 1$ and $a_{n+1} = 2a_n + 5$, $n=1,2,\dots$, then a_{100} is equal to:

- A $(5 * 2^{99} - 6)$
- B $(5 * 2^{99} + 6)$
- C $(6 * 2^{99} + 5)$
- D $(6 * 2^{99} - 5)$

cracku

If $a_1 = 1$ and $a_{n+1} = 2a_n + 5$, $n=1, 2, \dots$, then a_{100} is equal to:

A) $(5 \times 2^{99} - 5)$
 B) $(5 \times 2^{99} + 5)$
 C) $(6 \times 2^{99} + 5)$
 D) $(6 \times 2^{99} - 5)$

Handwritten solution:

$$a_1 = 1$$

$$a_2 = 2 \cdot 1 + 5 = 7$$

$$a_3 = 2(2+5) + 5 = 2^2 + 2 \times 5 + 5 = 2^2 + 3 \times 5$$

$$a_4 = 2(2^2 + 3 \times 5) + 5 = 2^3 + 3 \times 5 + 5 = 2^3 + 4 \times 5$$

$$a_5 = 2(2^3 + 4 \times 5) + 5 = 2^4 + 4 \times 5 + 5 = 2^4 + 5 \times 5$$

Pattern: $a_n = 2^n + n \times 5$

For $n=100$: $a_{100} = 2^{100} + 100 \times 5 = 2^{100} + 500$

None of the options match the derived formula.

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CAT Formula PDF

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Question 16

What is the value of the following expression?

$$\frac{1}{(2^2 - 1)} + \frac{1}{(4^2 - 1)} + \frac{1}{(6^2 - 1)} + \dots + \frac{1}{(20^2 - 1)}$$

- A 9/19
- B 10/19
- C 10/21
- D 11/21

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What is the value of the following expression? $\frac{1}{(2^2 - 1)} + \frac{1}{(4^2 - 1)} + \frac{1}{(6^2 - 1)} + \dots + \frac{1}{(20^2 - 1)}$

A) 9/19
 B) 10/19
 C) 10/21
 D) 11/21

Handwritten solution:

$$S = \frac{1}{2^2 - 1} + \frac{1}{4^2 - 1} + \frac{1}{6^2 - 1} + \dots + \frac{1}{20^2 - 1}$$

$$= \frac{1}{(2-1)(2+1)} + \frac{1}{(4-1)(4+1)} + \frac{1}{(6-1)(6+1)} + \dots + \frac{1}{(20-1)(20+1)}$$

$$= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{19 \times 21}$$

Pattern: $\frac{1}{(n-1)(n+1)} = \frac{1}{2} \left(\frac{1}{n-1} - \frac{1}{n+1} \right)$

For $n=2$ to $n=20$:

$$S = \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \dots + \frac{1}{19} - \frac{1}{21} \right)$$

$$= \frac{1}{2} \left(1 - \frac{1}{21} \right) = \frac{1}{2} \times \frac{20}{21} = \frac{10}{21}$$

VIDEO SOLUTION

Question 17

The n th element of a series is represented as

$$X_n = (-1)^n X_{n-1}$$

If $X_0 = x$ and $x > 0$, then which of the following is always true?

- A X_n is positive if n is even

- B X_n is positive if n is odd
- C X_n is negative if n is even
- D None of these

[VIEW SOLUTION](#)

Question 18

Let S denotes the infinite sum $2 + 5x + 9x^2 + 14x^3 + 20x^4 + \dots$, where $|x| < 1$ and the coefficient of x^{n-1} is $n(n+3)/2$, ($n = 1, 2, \dots$). Then S equals:

- A $(2-x)/(1-x)^3$
- B $(2-x)/(1+x)^3$
- C $(2+x)/(1-x)^3$
- D $(2+x)/(1+x)^3$

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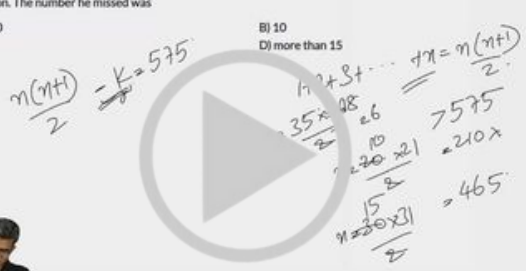
Question 19

A child was asked to add first few natural numbers (i.e. $1 + 2 + 3 + \dots$) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was

- A less than 10
- B 10
- C 15
- D more than 15

A child was asked to add first few natural numbers (i.e. $1 + 2 + 3 + \dots$) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was

A) less than 10 B) 10
C) 15 D) more than 15



[VIDEO SOLUTION](#)

Question 20

What is the sum of the following series: $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} \dots + \frac{1}{100 \times 101}$?

- A $\frac{99}{100}$
- B $\frac{1}{100}$
- C $\frac{100}{101}$
- D $\frac{101}{102}$

The video shows a teacher explaining the sum of the series $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{100 \times 101}$. He uses the method of partial fractions, showing that $\frac{1}{n \times (n+1)} = \frac{1}{n} - \frac{1}{n+1}$. He then writes out the first few terms of the series, showing how they telescope: $\frac{1}{1 \times 2} = \frac{1}{1} - \frac{1}{2}$, $\frac{1}{2 \times 3} = \frac{1}{2} - \frac{1}{3}$, $\frac{1}{3 \times 4} = \frac{1}{3} - \frac{1}{4}$, and so on, up to $\frac{1}{100 \times 101} = \frac{1}{100} - \frac{1}{101}$. The final sum is $1 - \frac{1}{101} = \frac{100}{101}$.

[VIDEO SOLUTION](#)

Question 21

The value of $\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$

- A $\frac{8}{1-x^8}$
- B $\frac{4x}{1+x^2}$
- C $\frac{4}{1-x^6}$
- D $\frac{4}{1+x^4}$

The video shows a teacher explaining the value of the expression $\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$. He starts by combining the first two terms: $\frac{1}{1-x} + \frac{1}{1+x} = \frac{1+x+1-x}{1-x^2} = \frac{2}{1-x^2}$. Then he adds the third term: $\frac{2}{1-x^2} + \frac{2}{1+x^2} = \frac{2(1+x^2) + 2(1-x^2)}{(1-x^2)(1+x^2)} = \frac{2(1+x^2+1-x^2)}{1-x^4} = \frac{4}{1-x^4}$. Finally, he adds the fourth term: $\frac{4}{1-x^4} + \frac{4}{1+x^4} = \frac{4(1+x^4) + 4(1-x^4)}{(1-x^4)(1+x^4)} = \frac{4(1+x^4+1-x^4)}{1-x^8} = \frac{8}{1-x^8}$.

[VIDEO SOLUTION](#)



Question 22

Consider a function $f(k)$ defined for positive integers $k = 1, 2, \dots$; the function satisfies the condition $f(1) + f(2) + \dots = \frac{p}{p-1}$. Where p is fraction i.e. $0 < p < 1$. Then $f(k)$ is given by

- A $(-p)(p)^{(k-1)}$
- B $p(1-p)^{k-1}$
- C $(p(1-p))^{k-1}$
- D None of these

[VIEW SOLUTION](#)

Question 23

N the set of natural numbers is partitioned into subsets $S_1 = (1)$, $S_2 = (2, 3)$, $S_3 = (4, 5, 6)$, $S_4 = (7, 8, 9, 10)$ and so on. The sum of the elements of the subset S_{50} is

- A 61250
- B 65525
- C 42455
- D 62525

The video solution shows the following calculations:

$$S_n = \frac{n(n+1)}{2}$$
$$S_{50} = \frac{50 \times 51}{2} = 51 \times 25 = 1275$$
$$S_{49} = \frac{49 \times 50}{2} = 49$$

[VIDEO SOLUTION](#)

Question 24

A square is drawn by joining the midpoints of the sides of a given square. A third square is drawn inside the second square in the same way and this process is continued indefinitely. If a side of the first square is 8 cm, the sum of the areas of all the squares such formed (in sq.cm.) is

- A 128
- B 120

- C 96
- D None of these

[VIEW SOLUTION](#)



Question 25

Three times the first of three consecutive odd integers is 3 more than twice the third. What is the third integer?

- A 15
- B 9
- C 11
- D 5

[VIEW SOLUTION](#)

Question 26

If the square of the 7th term of an arithmetic progression with positive common difference equals the product of the 3rd and 17th terms, then the ratio of the first term to the common difference is

- A 2:3
- B 3:2
- C 3:4
- D 4:3

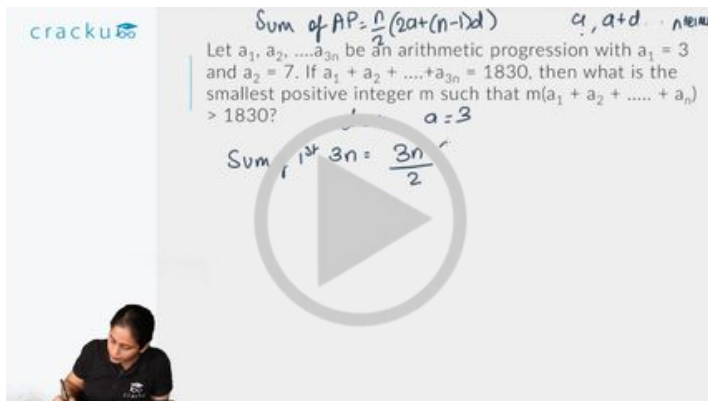
A video player interface for the solution to Question 26. It features a large play button in the center. Handwritten notes are visible on the left side of the video frame, showing the derivation of the ratio. The notes include: 'a', 'd > 0', '7th term = a + 6d', '3rd term = a + 2d', and '17th term = a + 16d'. The video title at the top reads: 'If the square of the 7th term of an arithmetic progression with positive common difference equals the product of the 3rd and 17th terms, then the ratio of the first term to the common difference is'. Below the title are the options: A) 2:3, B) 3:2, C) 3:4, D) 4:3. A small inset video shows a person writing at a desk.

[VIDEO SOLUTION](#)

Question 27

Let $a_1, a_2, \dots, a_{3n}$ be an arithmetic progression with $a_1 = 3$ and $a_2 = 7$. If $a_1 + a_2 + \dots + a_{3n} = 1830$, then what is the smallest positive integer m such that $m(a_1 + a_2 + \dots + a_n) > 1830$?

- A 8
- B 9
- C 10
- D 11



Sum of AP = $\frac{n}{2}(2a + (n-1)d)$ $a, a+d, \dots, n \text{ terms}$

Let $a_1, a_2, \dots, a_{3n}$ be an arithmetic progression with $a_1 = 3$ and $a_2 = 7$. If $a_1 + a_2 + \dots + a_{3n} = 1830$, then what is the smallest positive integer m such that $m(a_1 + a_2 + \dots + a_n) > 1830$?

$a = 3$

Sum of 1st $3n = \frac{3n}{2}$

VIDEO SOLUTION



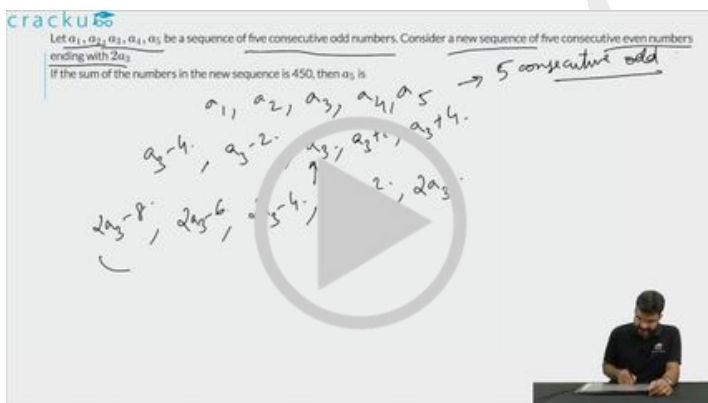
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Question 28

Let a_1, a_2, a_3, a_4, a_5 be a sequence of five consecutive odd numbers. Consider a new sequence of five consecutive even numbers ending with $2a_3$

If the sum of the numbers in the new sequence is 450, then a_5 is



Let a_1, a_2, a_3, a_4, a_5 be a sequence of five consecutive odd numbers. Consider a new sequence of five consecutive even numbers ending with $2a_3$. If the sum of the numbers in the new sequence is 450, then a_5 is

$a_1, a_2, a_3, a_4, a_5 \rightarrow 5 \text{ consecutive odd}$

$a_3 - 4, a_3 - 2, a_3, a_3 + 2, a_3 + 4$

$2a_3 - 8, 2a_3 - 6, 2a_3, 2a_3 + 2, 2a_3 + 4$

VIDEO SOLUTION

Question 29

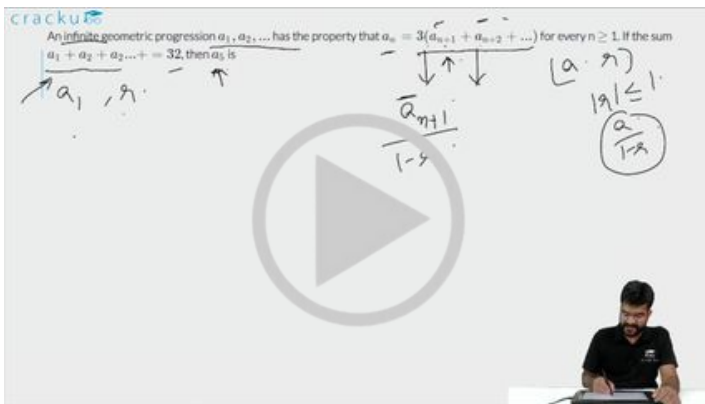
An infinite geometric progression $a_1, a_2, \dots$ has the property that $a_n = 3(a_{n+1} + a_{n+2} + \dots)$ for every $n \geq 1$.

If the sum $a_1 + a_2 + a_3 + \dots = 32$, then a_5 is

- A $1/32$
- B $2/32$

C $3/32$

D $4/32$



VIDEO SOLUTION

Question 30

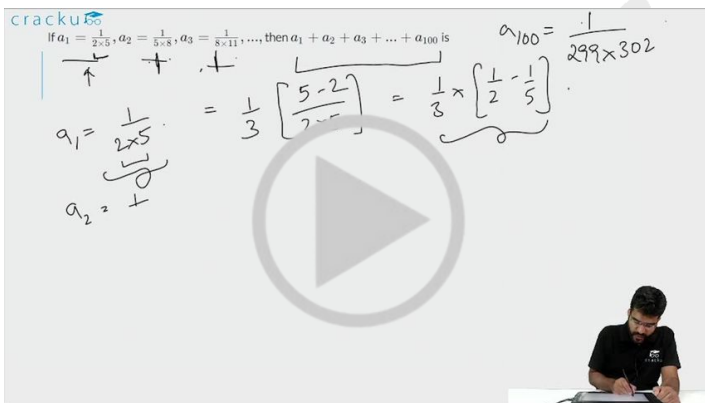
If $a_1 = \frac{1}{2 \times 5}$, $a_2 = \frac{1}{5 \times 8}$, $a_3 = \frac{1}{8 \times 11}$, ..., then $a_1 + a_2 + a_3 + \dots + a_{100}$ is

A $\frac{25}{151}$

B $\frac{1}{2}$

C $\frac{1}{4}$

D $\frac{111}{55}$



VIDEO SOLUTION



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Question 31

Let x, y, z be three positive real numbers in a geometric progression such that $x < y < z$. If $5x, 16y$, and $12z$ are in an arithmetic progression then the common ratio of the geometric progression is

A $\frac{3}{6}$

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The value of the sum $7 \times 11 + 11 \times 15 + 15 \times 19 + \dots + 95 \times 99$ is

$$S = 7 \times 11 + 11 \times 15 + 15 \times 19 + \dots + 95 \times 99$$

$$S = \sum_{n=1}^{23} (4n+3) \times (4n+7)$$

$$4n+3 = 95 \Rightarrow n = 23$$

VIDEO SOLUTION

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Step by step, heading towards 99+ percentile

Question 34

The arithmetic mean of x, y and z is 80, and that of x, y, z, u and v is 75, where $u = (x+y)/2$ and $v = (y+z)/2$. If $x \geq z$, then the minimum possible value of x is

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The arithmetic mean of x, y and z is 80, and that of x, y, z, u and v is 75, where $u = (x+y)/2$ and $v = (y+z)/2$. If $x \geq z$, then the minimum possible value of x is

$$\frac{x+y+z}{3} = 80 \Rightarrow x+y+z = 240$$

$$\frac{x+y+z+u+v}{5} = 75 \Rightarrow x+y+z+u+v = 375$$

$$u = \frac{x+y}{2} \quad v = \frac{y+z}{2}$$

$$x+y+z + \frac{x+y}{2} + \frac{y+z}{2} = 375$$

$$x+y+z + \frac{x+y+z}{2} = 375$$

$$\frac{3(x+y+z)}{2} = 375$$

$$x+y+z = 250$$

$$240 = 250 - z \Rightarrow z = 10$$

$$x \geq z \Rightarrow x \geq 10$$

VIDEO SOLUTION

Question 35

Let $t_1, t_2, \dots$ be real numbers such that $t_1 + t_2 + \dots + t_n = 2n^2 + 9n + 13$, for every positive integer $n \geq 2$. If $t_k = 103$, then k equals

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Let $t_1, t_2, \dots$ be real numbers such that $t_1 + t_2 + \dots + t_n = 2n^2 + 9n + 13$, for every positive integer $n \geq 2$. If $t_k = 103$, then k equals

$$t_1 + t_2 + t_3 + \dots + t_{n-1} + t_n = 2n^2 + 9n + 13$$

$$t_1 + t_2 + t_3 + \dots + t_{n-1} = 2(n-1)^2 + 9(n-1) + 13$$

$$t_n = (2n^2 + 9n + 13) - (2(n-1)^2 + 9(n-1) + 13)$$

$$t_n = 2n^2 + 9n + 13 - (2n^2 - 4n + 2 + 9n - 9 + 13)$$

$$t_n = 2n^2 + 9n + 13 - (2n^2 + 5n + 10)$$

$$t_n = 4n + 3$$

$$4k + 3 = 103 \Rightarrow k = 25$$

VIDEO SOLUTION

Question 36

Let $a_1, a_2, \dots$ be integers such that

$$a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n-1} a_n = n, \text{ for all } n \geq 1.$$

Then $a_{51} + a_{52} + \dots + a_{1023}$ equals

- A 0
- B 1
- C 10
- D -1

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Let $a_1, a_2, \dots$ be integers such that
 $a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n-1} a_n = n$, for all $n \geq 1$.
Then $a_{51} + a_{52} + \dots + a_{1023}$ equals

$a_1 = 1$
 $a_1 - a_2 = 2 \Rightarrow a_2 = -1$
 $a_1 - a_2 + a_3 = 3 \Rightarrow a_3 = 1$
 $a_1 - a_2 + a_3 - a_4 = 4 \Rightarrow a_4 = -1$

Then $a_{51} + a_{52} + \dots + a_{1023} = -1$

VIDEO SOLUTION

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Question 37

If $(2n + 1) + (2n + 3) + (2n + 5) + \dots + (2n + 47) = 5280$, then what is the value of $1 + 2 + 3 + \dots + n$?

cracku

If $(2n + 1) + (2n + 3) + (2n + 5) + \dots + (2n + 47) = 5280$, then what is the value of $1 + 2 + 3 + \dots + n$?

$n = 24$

$1, 3, 5, \dots, 47$

$\frac{47 - 1}{2} + 1 = 24$

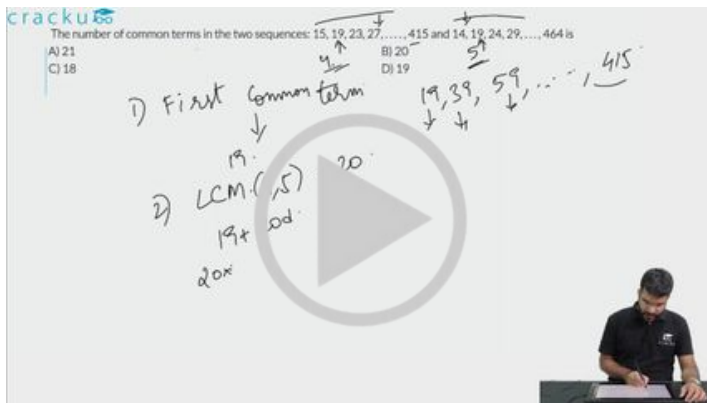
$24 \times 24 + (1 + 3 + \dots + 47) = 5280$

VIDEO SOLUTION

Question 38

The number of common terms in the two sequences: 15, 19, 23, 27, . . . , 415 and 14, 19, 24, 29, . . . , 464 is

- A 21
- B 20
- C 18
- D 19



VIDEO SOLUTION

Question 39

If $a_1, a_2, \dots$ are in A.P., then, $\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_n} + \sqrt{a_{n+1}}}$ is equal to

- A $\frac{n}{\sqrt{a_1} + \sqrt{a_{n+1}}}$
- B $\frac{n-1}{\sqrt{a_1} + \sqrt{a_{n-1}}}$
- C $\frac{n-1}{\sqrt{a_1} + \sqrt{a_n}}$
- D $\frac{n}{\sqrt{a_1} - \sqrt{a_{n+1}}}$



VIDEO SOLUTION



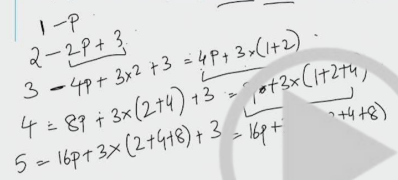
Question 40

If the population of a town is p in the beginning of any year then it becomes $3 + 2p$ in the beginning of the next year. If the population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be

- A $(1003)^{15} + 6$
- B $(997)^{15} - 3$
- C $(997)2^{14} + 3$
- D $(1003)2^{15} - 3$

cracku

If the population of a town is p in the beginning of any year then it becomes $3 + 2p$ in the beginning of the next year.
If the population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be



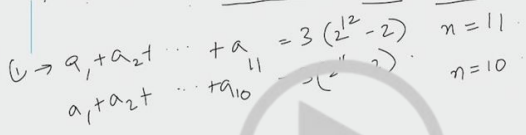
VIDEO SOLUTION

Question 41

If $a_1 + a_2 + a_3 + \dots + a_n = 3(2^{n+1} - 2)$, for every $n \geq 1$, then a_{11} equals

cracku

If $a_1 + a_2 + a_3 + \dots + a_n = 3(2^{n+1} - 2)$, for every $n \geq 1$, then a_{11} equals



VIDEO SOLUTION

Question 42

If $x_1 = -1$ and $x_m = x_{m+1} + (m + 1)$ for every positive integer m , then X_{100} equals

- A -5050
- B -5151
- C -5051
- D -5150

If $x_1 = -1$ and $x_m = x_{m+1} + (m+1)$ for every positive integer m , then X_{100} equals

A) -5050
B) -5151
C) -5051
D) -5150

$x_1 = -1$
 $x_{m+1} = x_m - (m+1)$
 $x_2 = x_1 - 2 = -1 - 2 = -3$
 $x_3 = x_2 - 3 = -3 - 3 = -6$

VIDEO SOLUTION



Question 43

Let the m -th and n -th terms of a geometric progression be $\frac{3}{4}$ and 12, respectively, where $m < n$. If the common ratio of the progression is an integer r , then the smallest possible value of $r + n - m$ is

- A 6
- B 2
- C -4
- D -2

Let the m -th and n -th terms of a geometric progression be $\frac{3}{4}$ and 12, respectively, where $m < n$. If the common ratio of the progression is an integer r , then the smallest possible value of $r + n - m$ is

$a \cdot r^{m-1} = \frac{3}{4}$
 $a \cdot r^{n-1} = 12$
 $\frac{a \cdot r^{n-1}}{a \cdot r^{m-1}} = \frac{12}{\frac{3}{4}}$
 $r^{n-m} = 16$
 $n-m = 4$

VIDEO SOLUTION

Question 44

If $x_0 = 1$, $x_1 = 2$, and $x_{n+2} = \frac{1+x_{n+1}}{x_n}$, $n = 0, 1, 2, 3, \dots$, then x_{2021} is equal to

- A 4
- B 1
- C 3

D 2

cracku

If $x_0 = 1, x_1 = 2$, and $x_{n+2} = \frac{1+x_{n+1}}{x_n}$, $n = 0, 1, 2, 3, \dots$, then x_{2021} is equal to

A) 4
B) 1
C) 3
D) 2

$x_0 = 1$
 $x_1 = 2$
 $x_2 = \frac{1+2}{1} = 3$
 $x_3 = \frac{1+3}{2} = 2$
 $x_4 = \frac{1+2}{3} = 1$
 $x_5 = \frac{1+1}{2} = 1$
 $x_6 = \frac{1+1}{1} = 2$
 $x_7 = \frac{1+2}{1} = 3$



VIDEO SOLUTION

Question 45

The natural numbers are divided into groups as (1), (2, 3, 4), (5, 6, 7, 8, 9), and so on. Then, the sum of the numbers in the 15th group is equal to

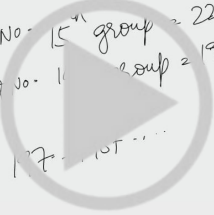
- A 6119
- B 6090
- C 4941
- D 7471

cracku

The natural numbers are divided into groups as (1), (2, 3, 4), (5, 6, 7, 8, 9), and so on. Then, the sum of the numbers in the 15th group is equal to

A) 6119
B) 6090
C) 4941
D) 7471

Last No. of 15th group = 225
Last No. of 14th group = 196
197, 198, ...



VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

Three positive integers x, y and z are in arithmetic progression. If $y - x > 2$ and $xyz = 5(x + y + z)$, then $z - x$ equals

- A 8

- B 12
- C 14
- D 10

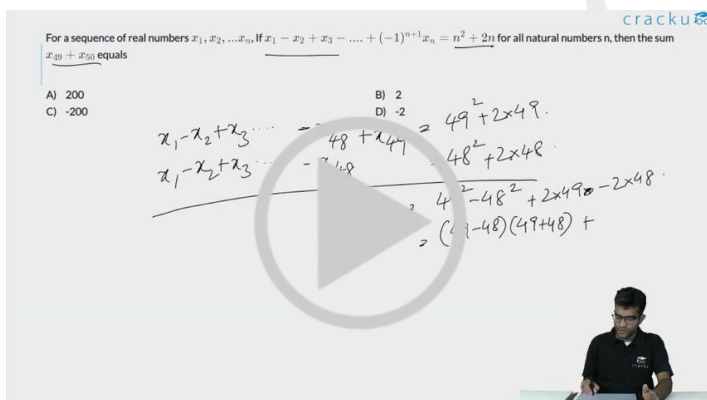


VIDEO SOLUTION

Question 47

For a sequence of real numbers $x_1, x_2, \dots, x_n$, if $x_1 - x_2 + x_3 - \dots + (-1)^{n+1}x_n = n^2 + 2n$ for all natural numbers n , then the sum $x_{49} + x_{50}$ equals

- A 200
- B 2
- C -200
- D -2



VIDEO SOLUTION

Question 48

Consider a sequence of real numbers, $x_1, x_2, x_3, \dots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

- A 4849
- B 4949

C 4950

D 4850

cracku

Consider a sequence of real numbers, $x_1, x_2, x_3, \dots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

A) 4849
B) 4949
C) 4950
D) 4850

$x_1 = -1$
 $x_2 = x_1 + 0$
 $x_3 = x_2 + 1$
 $x_4 = x_3 + 2$
 $\vdots$
 $x_{100} = x_{99} + 99$
 $x_1 + x_2 + \dots + x_{100} = x_1 + \dots$

VIDEO SOLUTION

100+ CAT DILR Questions



Question 49

For any natural number n , suppose the sum of the first n terms of an arithmetic progression is $(n + 2n^2)$. If the n^{th} term of the progression is divisible by 9, then the smallest possible value of n is

A 9

B 4

C 7

D 8

cracku

For any natural number n , suppose the sum of the first n terms of an arithmetic progression is $(n + 2n^2)$. If the n^{th} term of the progression is divisible by 9, then the smallest possible value of n is

A) 9
B) 4
C) 7
D) 8

$T_1 + T_2 + \dots + T_{n-1} + T_n$
 S_{n-1}
 S_n
 T_n
 $S_n = 2n^2 + n$
 $S_{n-1} = 2(n-1)^2 + (n-1)$

VIDEO SOLUTION

Question 50

The average of a non-decreasing sequence of N numbers $a_1, a_2, \dots, a_N$ is 300. If a_1 is replaced by $6a_1$, the new average becomes 400. Then, the number of possible values of a_1 , is

The average of a non-decreasing sequence of N numbers $a_1, a_2, \dots, a_N$ is 300. If a_1 is replaced by $6a_1$, the new average becomes 400. Then, the number of possible values of a_1 is _____

Answer: _____

$a_1 + a_2 + \dots + a_N = 300N$
 $6a_1 + a_2 + \dots + a_N = 400N$
 $5a_1 = 100N$
 $a_1 = 20N$
 $a_1 \leq 300$
 $N = 1, 2, 3, \dots, 15$
 $a_1 = 20, 40, 60, \dots, 300$

VIDEO SOLUTION

Question 51

On day one, there are 100 particles in a laboratory experiment. On day n , where $n \geq 2$, one out of every n articles produces another particle. If the total number of particles in the laboratory experiment increases to 1000 on day m , then m equals

- A 19
- B 16
- C 18
- D 17

On day one, there are 100 particles in a laboratory experiment. On day n , where $n \geq 2$, one out of every n articles produces another particle. If the total number of particles in the laboratory experiment increases to 1000 on day m , then m equals

A) 19
B) 16
C) 18
D) 17

Day 1 $\rightarrow 100$
 Day 2 $\rightarrow \frac{100}{2} = 50$
 Day 3 $\rightarrow \frac{150}{3} = 50$
 150

VIDEO SOLUTION

100+ CAT VARC Questions



Question 52

Consider the arithmetic progression 3, 7, 11, ... and let A_n denote the sum of the first n terms of this progression. Then the value of $\frac{1}{25} \sum_{n=1}^{25} A_n$ is

- A 455
- B 442

C 415

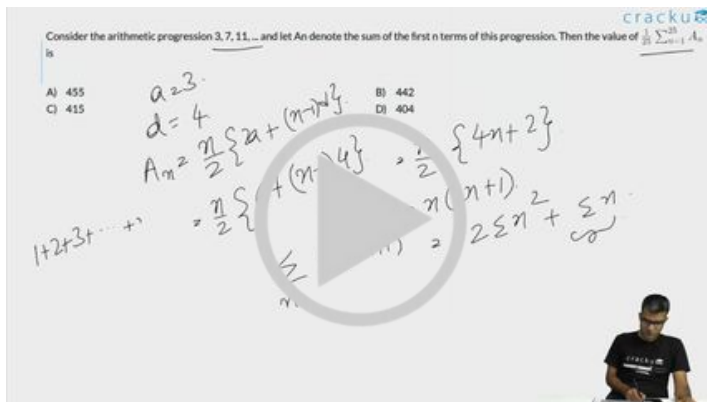
D 404

Consider the arithmetic progression 3, 7, 11, ... and let A_n denote the sum of the first n terms of this progression. Then the value of $\sum_{n=1}^{\infty} \frac{A_n}{2^n}$ is

A) 455
C) 415

B) 442
D) 404

$a = 3$
 $d = 4$
 $A_n = \frac{n}{2} \{2a + (n-1)d\} = \frac{n}{2} \{2 \cdot 3 + (n-1) \cdot 4\} = \frac{n}{2} \{4n + 2\} = \frac{n}{2} (4n + 2)$
 $1+2+3+\dots+n = \frac{n(n+1)}{2}$
 $\sum_{n=1}^{\infty} \frac{A_n}{2^n} = \sum_{n=1}^{\infty} \frac{n(4n+2)}{2^{n+1}} = \sum_{n=1}^{\infty} \frac{n(4n+2)}{2^{n+1}}$



VIDEO SOLUTION

Question 53

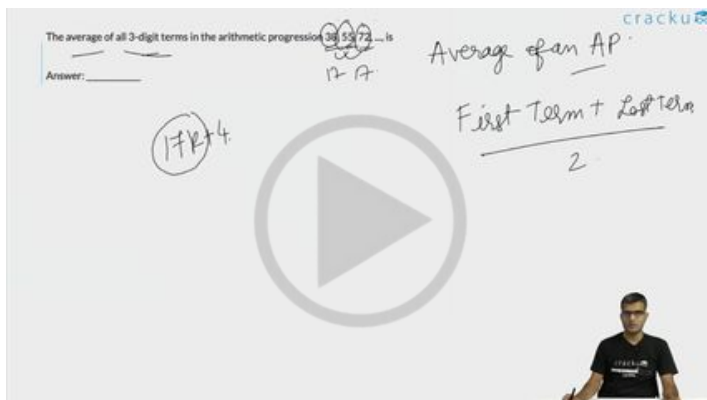
The average of all 3-digit terms in the arithmetic progression 38, 55, 72, ..., is

The average of all 3-digit terms in the arithmetic progression 38, 55, 72, ... is

Answer: _____

Average of an AP:
 $\frac{\text{First Term} + \text{Last Term}}{2}$

(17×4)



VIDEO SOLUTION

Question 54

For some positive and distinct real numbers x, y and z , if $\frac{1}{\sqrt{y} + \sqrt{z}}$ is the arithmetic mean of $\frac{1}{\sqrt{x} + \sqrt{z}}$ and $\frac{1}{\sqrt{x} + \sqrt{y}}$, then the relationship which will always hold true, is

- A $\sqrt{x}, \sqrt{z}$ and $\sqrt{y}$ are in arithmetic progression
- B y, x and z are in arithmetic progression
- C x, y and z are in arithmetic progression
- D $\sqrt{x}, \sqrt{y}$ and $\sqrt{z}$ are in arithmetic progression

 VIDEO SOLUTION

 VIDEO SOLUTION

- A** $\log_4 \left(\frac{3}{2} \right)$
B $\log_4 7$
C $\log_4 \left(\frac{23}{2} \right)$
D $\log_4 \left(\frac{7}{2} \right)$

For a real number x , if $\frac{\log(2^x-9)}{\log 4}$, $\frac{\log(2^x-9)}{\log 4}$, and $\frac{\log(2^x-9)}{\log 4}$ are in an arithmetic progression, then the common difference is

A) $\log_4\left(\frac{3}{2}\right)$
 B) $\log_4 7$
 C) $\log_4\left(\frac{2}{3}\right)$
 D) $\log_4\left(\frac{1}{2}\right)$

Handwritten notes show the progression of terms and the use of logarithmic properties to solve for the common difference.

VIDEO SOLUTION

Question 57

The value of $1 + \left(1 + \frac{1}{3}\right) \frac{1}{4} + \left(1 + \frac{1}{3} + \frac{1}{9}\right) \frac{1}{16} + \left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27}\right) \frac{1}{64} + \dots$ is

- A $\frac{15}{13}$
 B $\frac{27}{12}$
 C $\frac{15}{8}$
 D $\frac{16}{11}$

The value of $1 + \left(1 + \frac{1}{3}\right) \frac{1}{4} + \left(1 + \frac{1}{3} + \frac{1}{9}\right) \frac{1}{16} + \left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27}\right) \frac{1}{64} + \dots$ is

A) $\frac{15}{13}$
 B) $\frac{27}{12}$
 C) $\frac{15}{8}$
 D) $\frac{16}{11}$

Handwritten notes show the summation of the series using the formula for the sum of a geometric series.

VIDEO SOLUTION

IMPORTANT **CAT Important Topics**

Question 58

Let $a_n = 46 + 8n$ and $b_n = 98 + 4n$ be two sequences for natural numbers $n \leq 100$. Then, the sum of all terms common to both the sequences is

- A 14900
 B 14798
 C 15000

D 14602

Let $a_n = 46 + 8n$ and $b_n = 98 + 4n$ be two sequences for natural numbers $n \leq 100$. Then, the sum of all terms common to both the sequences is

A) 14900 B) 14798
C) 15000 D) 14602

VIDEO SOLUTION

Question 59

Let both the series $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be in arithmetic progression such that the common differences of both the series are prime numbers. If $a_5 = b_9$, $a_{19} = b_{19}$ and $b_2 = 0$, then a_{11} equals

- A 86
- B 79
- C 83
- D 84

Let both the series $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be in arithmetic progression such that the common differences of both the series are prime numbers. If $a_5 = b_9$, $a_{19} = b_{19}$ and $b_2 = 0$, then a_{11} equals

A) 86 B) 79
C) 83 D) 84

VIDEO SOLUTION

Question 60

Let a_n and b_n be two sequences such that $a_n = 13 + 6(n - 1)$ and $b_n = 15 + 7(n - 1)$ for all natural numbers n . Then, the largest three digit integer that is common to both these sequences, is

Let a_n and b_n be two sequences such that $a_n = 13 + (n-1)$ and $b_n = 15 + (n-1)$ for all natural numbers n . Then, the largest three digit integer that is common to both these sequences, is

Answer: _____

Handwritten notes show two arithmetic sequences:

Sequence 1: 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

Sequence 2: 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

The common terms are 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

The largest three digit integer common to both sequences is 43.

VIDEO SOLUTION

CAT WhatsApp Group

cracku

WhatsApp icon

Question 61

The sum of the infinite series $\frac{1}{5} \left(\frac{1}{5} - \frac{1}{7} \right) + \left(\frac{1}{5} \right)^2 \left(\left(\frac{1}{5} \right)^2 - \left(\frac{1}{7} \right)^2 \right) + \left(\frac{1}{5} \right)^3 \left(\left(\frac{1}{5} \right)^3 - \left(\frac{1}{7} \right)^3 \right) + \dots$ is equal to

- A $\frac{7}{816}$
- B $\frac{5}{408}$
- C $\frac{7}{408}$
- D $\frac{5}{816}$

The sum of the infinite series $\frac{1}{5} \left(\frac{1}{5} - \frac{1}{7} \right) + \left(\frac{1}{5} \right)^2 \left(\left(\frac{1}{5} \right)^2 - \left(\frac{1}{7} \right)^2 \right) + \left(\frac{1}{5} \right)^3 \left(\left(\frac{1}{5} \right)^3 - \left(\frac{1}{7} \right)^3 \right) + \dots$ is equal to

Handwritten notes show the series terms: $a_1, a_2, a_3, a_4, \dots$

Options: A) $\frac{7}{816}$, B) $\frac{5}{408}$, C) $\frac{7}{408}$, D) $\frac{5}{816}$

Handwritten calculations show the sum of the series:

$\left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^4 + \left(\frac{1}{5} \right)^6 + \dots$

$-\frac{1}{5} \times \frac{1}{7} - \left(\frac{1}{5} \times \frac{1}{7} \right)^3 - \dots$

VIDEO SOLUTION

Question 62

Suppose $x_1, x_2, x_3, \dots, x_{100}$ are in arithmetic progression such that $x_5 = -4$ and $2x_6 + 2x_9 = x_{11} + x_{13}$. Then, x_{100} equals

- A -194
- B -196
- C 204
- D 206

The video shows a hand-drawn solution on a whiteboard. The text on the board reads: "Suppose $x_1, x_2, x_3, \dots, x_{100}$ are in arithmetic progression such that $x_5 = -4$ and $2x_6 + 2x_9 = x_{11} + x_{13}$. Then, x_{100} equals". Below this, the options are listed: A) -194, B) -196, C) 204, D) 206. The solution starts with $x_5 = -4 + 4d$, which simplifies to $-4 = -4 + 4d$, leading to $d = 1$. Then, $x_6 = -4 + d = -3$, $x_9 = -4 + 5d = 1$, $x_{11} = -4 + 7d = 3$, and $x_{13} = -4 + 9d = 5$. Substituting these into the equation $2x_6 + 2x_9 = x_{11} + x_{13}$, we get $2(-3) + 2(1) = 3 + 5$, which simplifies to $-6 + 2 = 8$, or $-4 = 8$, which is a contradiction. The correct approach is to use the general term $x_n = a + (n-1)d$. From $x_5 = -4$, we have $a + 4d = -4$. From $2x_6 + 2x_9 = x_{11} + x_{13}$, we have $2(a + 5d) + 2(a + 8d) = (a + 10d) + (a + 12d)$, which simplifies to $2a + 10d + 2a + 16d = 2a + 22d$, or $4a + 26d = 2a + 22d$, leading to $2a + 4d = 0$, or $a = -2d$. Substituting $a = -2d$ into $a + 4d = -4$, we get $-2d + 4d = -4$, or $2d = -4$, so $d = -2$. Then, $a = -2(-2) = 4$. Finally, $x_{100} = a + 99d = 4 + 99(-2) = 4 - 198 = -194$.

VIDEO SOLUTION

Question 63

For any natural number n let a_n be the largest integer not exceeding $\sqrt{n}$. Then the value of $a_1 + a_2 + \dots + a_{50}$ is

The video shows a hand-drawn solution on a whiteboard. The text on the board reads: "For any natural number n let a_n be the largest integer not exceeding $\sqrt{n}$. Then the value of $a_1 + a_2 + \dots + a_{50}$ is". Below this, the options are listed: A) 10, B) 15, C) 20, D) 25. The solution starts by listing the values of a_n for n from 1 to 50. For n from 1 to 3, $a_n = 1$. For n from 4 to 8, $a_n = 2$. For n from 9 to 15, $a_n = 3$. For n from 16 to 24, $a_n = 4$. For n from 25 to 35, $a_n = 5$. For n from 36 to 48, $a_n = 6$. For n from 49 to 50, $a_n = 7$. The sum is then calculated as $3(1) + 5(2) + 7(3) + 9(4) + 11(5) + 13(6) + 2(7) = 3 + 10 + 21 + 36 + 55 + 78 + 14 = 217$.

VIDEO SOLUTION

CAT Mock Test
Mocks designed exactly like the actual CAT

Question 64

Consider the sequence $t_1 = 1, t_2 = -1$ and $t_n = \left(\frac{n-3}{n-1}\right) t_{n-2}$ for $n \geq 3$. Then, the value of the sum

$$\frac{1}{t_2} + \frac{1}{t_4} + \frac{1}{t_6} + \dots + \frac{1}{t_{2022}} + \frac{1}{t_{2024}}, \text{ is}$$

- A -1024144
- B -1022121
- C -1023132
- D -1026169

The video shows a teacher explaining the sequence $t_1 = 1, t_2 = -1$ and $t_n = \left(\frac{n-3}{n-1}\right) t_{n-2}$ for $n \geq 3$. The teacher calculates the first few terms: $t_3 = \frac{0}{2} \times 1 = 0$, $t_4 = \frac{1}{3} \times t_2 = -\frac{1}{3}$, $t_5 = \frac{0}{4} \times t_3 = 0$, $t_6 = \frac{1}{5} \times t_4 = -\frac{1}{5}$, and $t_7 = 0$. The teacher then discusses the sum $\frac{1}{t_2} + \frac{1}{t_4} + \frac{1}{t_6} + \dots + \frac{1}{t_{2022}} + \frac{1}{t_{2024}}$.

VIDEO SOLUTION

Question 65

For any natural number k , let $a_k = 3^k$. The smallest natural number m for which $\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} < \{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$, is

- A 58
- B 59
- C 56
- D 57

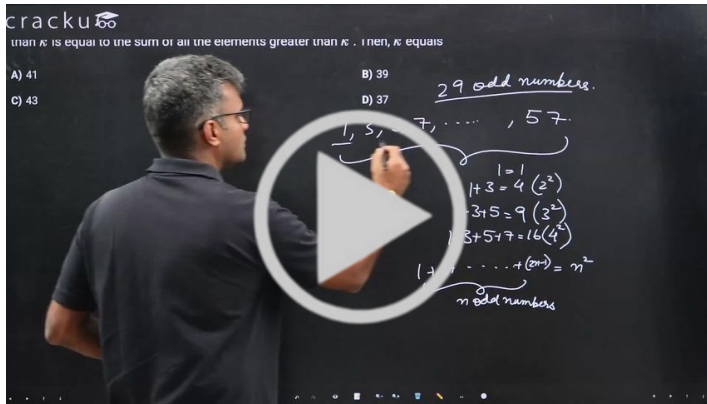
The video shows a teacher writing on a blackboard. The teacher is calculating the sum $1 + 4 + 9 + \dots + 400$ and the product $1 \times 2 \times 3 \times \dots \times 20$.

VIDEO SOLUTION

Question 66

In the set of consecutive odd numbers $\{1, 3, 5, \dots, 57\}$, there is a number k such that the sum of all the elements less than k is equal to the sum of all the elements greater than k . Then, k equals

- A 41
- B 39
- C 43
- D 37



[VIDEO SOLUTION](#)



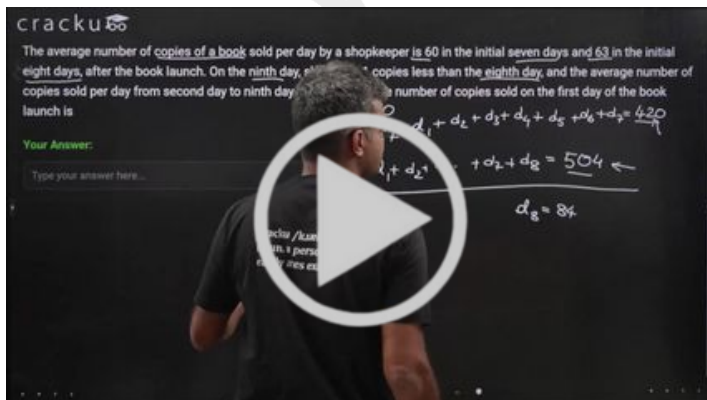
Question 67

In an arithmetic progression, if the sum of fourth, seventh and tenth terms is 99, and the sum of the first fourteen terms is 497, then the sum of first five terms is

[VIEW SOLUTION](#)

Question 68

The average number of copies of a book sold per day by a shopkeeper is 60 in the initial seven days and 63 in the initial eight days, after the book launch. On the ninth day, she sells 11 copies less than the eighth day, and the average number of copies sold per day from second day to ninth day becomes 66. The number of copies sold on the first day of the book launch is

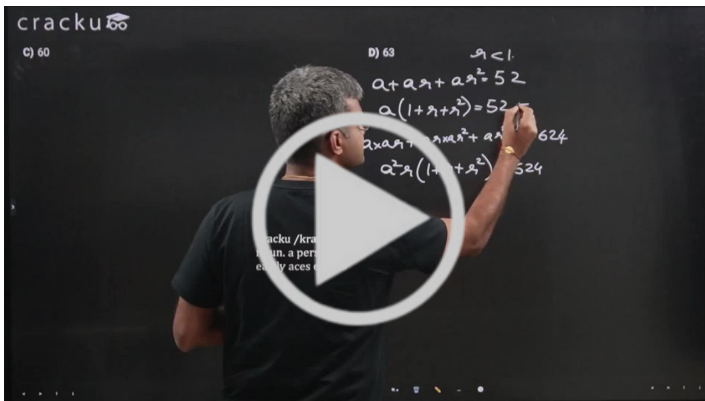


 VIDEO SOLUTION

Question 69

Let a_n be the n^{th} term of a decreasing infinite geometric progression. If $a_1 + a_2 + a_3 = 52$ and $a_1a_2 + a_2a_3 + a_3a_1 = 624$, then the sum of this geometric progression is

- A 57
- B 54
- C 60
- D 63



 VIDEO SOLUTION



Instructions

Answer the questions based on the following information. A series S_1 of five positive integers is such that the third term is half the first term and the fifth term is 20 more than the first term. In series S_2 , the n th term is defined as the difference between the $(n+1)$ th term and the n th term of series S_1 , S_2 is an arithmetic progression with a common difference of 30.

Question 70

First term of S_1 is

- A 80
- B 90
- C 100
- D 120

 VIEW SOLUTION

Question 71

Fourth term of S_2

- A 50
- B 60
- C 70
- D None of these

[VIEW SOLUTION](#)

Question 72

What is the difference between fourth terms of S_1 and S_2 ?

- A 10
- B 20
- C 30
- D 40

[VIEW SOLUTION](#)



CAT Syllabus PDF



Question 73

What is the average value of the terms of series S_1 ?

- A 60
- B 70
- C 80
- D Average is not an integer

[VIEW SOLUTION](#)

Question 74

What is the sum of series S_2 ?

- A 10
- B 20
- C 30
- D 40

[VIEW SOLUTION](#)

Instructions

DIRECTIONS for the following questions: These questions are based on the situation given below: There are fifty integers $a_1, a_2, \dots, a_{50}$, not all of them necessarily different. Let the greatest integer of these fifty integers be referred to as G , and the smallest integer be referred to as L . The integers a_1 through a_{24} form sequence $S1$, and the rest form sequence $S2$. Each member of $S1$ is less than or equal to each member of $S2$.

Question 75

All values in $S1$ are changed in sign, while those in $S2$ remain unchanged. Which of the following statements is true?

- A Every member of $S1$ is greater than or equal to every member of $S2$.
- B G is in $S1$
- C If all numbers originally in $S1$ and $S2$ had the same sign, then after the change of sign, the largest number of $S1$ and $S2$ is in $S1$.
- D None of the above

 [VIEW SOLUTION](#)



CAT Formula PDF



Question 76

Elements of $S1$ are in ascending order, and those of $S2$ are in descending order. a_{24} and a_{25} are interchanged. Then, which of the following statements is true?

- A $S1$ continues to be in ascending order
- B $S2$ continues to be in descending order
- C $S1$ continues to be in ascending order and $S2$ in descending order.
- D None of the above

 [VIEW SOLUTION](#)

Question 77

Every element of $S1$ is made greater than or equal to every element of $S2$ by adding to each element of $S1$ an integer x . Then x cannot be less than:

- A 2^{10}
- B The smallest value of $S2$
- C The largest value of $S2$
- D $(G-L)$

 [VIEW SOLUTION](#)

Answers

1.C	2.C	3.D	4.C	5.C	6.A	7.A	8.A
9.C	10.A	11.D	12.C	13.A	14.C	15.D	16.C
17.D	18.A	19.D	20.C	21.A	22.A	23.D	24.A
25.A	26.A	27.B	28.51	29.C	30.A	31.C	32.C
33.80707	34.105	35.24	36.B	37.4851	38.B	39.A	40.D
41.6144	42.A	43.D	44.D	45.A	46.C	47.D	48.D
49.C	50.14	51.A	52.A	53.548	54.B	55.19	56.D
57.D	58.A	59.B	60.967	61.B	62.A	63.217	64.A
65.A	66.A	67.65	68.49	69.B	70.C	71.A	72.B
73.C	74.B	75.D	76.A	77.D			

Explanations

1.C

The terms of the first sequence are of the form $4p + 13$

The terms of the second sequence are of the form $5q + 11$

If a term is common to both the sequences, it is of the form $4p+13$ and $5q+11$

or $4p = 5q - 2$. LHS = $4p$ is always even, so, q is also even.

or $2p = 5r - 1$ where $q = 2r$.

Notice that LHS is again even, hence r should be odd. Let $r = 2m+1$ for some m .

Hence, $p = 5m + 2$.

So, the number = $4p+13 = 20m + 21$.

Hence, all numbers of the form $20m + 21$ will be the common terms. i.e $21, 41, 61, \dots, 401 = 20$.

[▶ VIDEO SOLUTION](#)

2.C

The sum of the 3rd and 15th terms is $a+2d+a+14d = 2a+16d$

The sum of the 6th, 11th and 13th terms is $a+5d+a+10d+a+12d = 3a+27d$

Since the two are equal, $2a+16d = 3a+27d \Rightarrow a+11d = 0$

So, the 12th term is 0

[▶ VIDEO SOLUTION](#)

3.D

1, 2, 3, 4, ..., n such that the sum is greater than 288

If $n = 24$, $n(n+1)/2 = 12 \times 25 = 300$

So, $n = 24$, i.e. the 24th letter in the alphabet is the letter at position 288 in the series

So, answer = x

[▶ VIDEO SOLUTION](#)

4.C

For the given problem ,

$$\sum n(n+1)/2 = 8436 \text{ which is}$$

$$\sum n^2/2 + \sum n/2 = 8436 \text{ which is equal to}$$

$$n*(n+1)(2n+1)/12 + n*(n+1)/4 = 8436, \text{ solving we get } n=36.$$

Solving the equation might be lengthy. you can substitute the values in the options to arrive at the answer.

[VIDEO SOLUTION](#)

5. C

Let the numbers be $a_1, a_2, \dots, a_n$.

Since the numbers are positive,

$$AM \geq GM$$

$$\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \geq (a_1 * a_2 * \dots * a_n)^{1/n}$$

$$a_1 + a_2 + a_3 + \dots + a_n \geq n$$

[VIDEO SOLUTION](#)




6. A

Let there only be 2 questions.

Thus there are $2^{2-1} = 2$ students who have done 1 or more questions wrongly, $2^{2-2} = 1$ students who have done all 2 questions wrongly.

Thus total number of wrong answers = $2 + 1 = 3 = 2^2 - 1$.

Now let there be 3 questions. Then $j = 1, 2, 3$

Number of students answering 1 or more questions incorrectly = 4

Number of students answering 2 or more questions incorrectly = 2

Number of students answering 3 or more questions incorrectly = 1

Total number of incorrect answers = $1(3) + (2-1)*2 + (4-2)*1 = 7 = 2^3 - 1$

According to the question, the total number of wrong answers = $4095 = 2^{12} - 1$.

Hence Option A.

[VIEW SOLUTION](#)

7. A

Sum of the first 11 terms = $11/2 (2a+10d)$

Sum of the first 19 terms = $19/2 (2a+18d)$

$$\Rightarrow 22a+110d = 38a+342d \Rightarrow 16a = -232d$$

$$\Rightarrow 2a = -232/8 d = -29d$$

Sum of the first 30 terms = $15(2a+29d) = 0$

[VIDEO SOLUTION](#)

8. A

According to given condition we have ,

$$n+6b = nr^6 \text{ and } b=10.5n ,$$

$$63n+n = nr^6$$

$$r^6 = 64$$

$$r = 2$$

 VIDEO SOLUTION

9. C

According to given conditions the terms are 81.33, -19, -100.33, -81.33, 19, 100.33, 81.33, -19,... Hence the series repeats after every 6 terms . Also summation of these 6 terms is 0. Hence summation is 60002 terms will we sum of first 2 terms which is 62.33.

 VIDEO SOLUTION

10. A

substituting 3,4...53 in the given function, we get

$$\begin{aligned} t_3 &= \frac{3}{5} \\ t_4 &= \frac{4}{6} \\ t_5 &= \frac{5}{7} \\ t_6 &= \frac{6}{8} \end{aligned}$$

Multiplying the values, we get $\frac{3}{5} * \frac{4}{6} * \frac{5}{7} * \dots * \frac{52}{54} * \frac{53}{55}$ which ultimately after cancellations give $\frac{3*4}{54*55} = \frac{2}{495}$

 VIDEO SOLUTION

CAT Percentile Predictor



11. D

Let x be in the front row.

So no. of children in next rows will be x-3,x-6,x-9,x-12,x-15,x-18,x-21....

Suppose there are 6 rows, then the sum is equal to $x + x-3 + x-6 + x-9 + x-12 + x-15 = 6x - 45$

This sum is equal to 630.

$$\Rightarrow 6x - 45 = 630 \Rightarrow 6x = 675$$

Here, x is not an integer.

Hence, there cannot be 6 rows.

 VIDEO SOLUTION

12. C

$$x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots}}}} \text{ to infinity}$$

$$\Rightarrow x = \sqrt{4 + \sqrt{4 - x}}$$

$$\Rightarrow x^2 = 4 + \sqrt{4 - x}$$

$$\Rightarrow x^4 + 16 - 8x^2 = 4 - x$$

$$\Rightarrow x^4 - 8x^2 + x + 12 = 0$$

On substituting options, we can see that option C satisfies the equation.

[VIEW SOLUTION](#)

13. A

January 1, 1950 to December 31, 1959 is a period of 10 years or 20 half years.

The person X after 1st year gets Rs. 300 in next year he gets Rs. 330 and so on.

So his earning is in AP with 10 $300+330+360+\dots$

Similarly earning of Y is in AP with 20 terms $200+215+230+245+\dots$

So, the total earnings of X equals $12 \times (300+330+\dots 10 \text{ terms}) = 52200$

The total earnings of Y equals $6 \times (200+215+230+\dots 20 \text{ terms}) = 41100$

So, the total earnings of the two equals $52200+41100 = 93300$

[VIDEO SOLUTION](#)

14. C

It is given that in a Fibonacci sequence, from the third term on wards, each term in the sequence is the sum of the previous two terms in that sequence.

Let x and y be the 1st and 2nd term respectively.

3rd term = x+y

4th term = x+2y

5th term = 2x+3y

6th term = 3x+5y

7th term = 5x+8y

We know that difference of the squares of 6th and 7th terms is $517 = 47 \times 11$.

And $a^2 - b^2 = (a + b)(a - b)$.

Applying above formula we get $(8x+13y)(2x+3y) = 47 \times 11$.

So only possible way is $(8x+13y)=47$ and

$2x+3y=11$.

Solving we get x=1 and y=3.

Using the concept that every term is the sum of the previous two terms, as used in the beginning of the solution, we get 10th term as $21x+34y$, which gives 10th term as 123.

[VIDEO SOLUTION](#)

15. D

$$a_2 = 2 \times 1 + 5$$

$$a_3 = 2 \times (2 + 5) + 5 = 2^2 + 5 \times 2 + 5$$

$$a_4 = 2^3 + 5 \times 2^2 + 5 \times 2 + 5$$

...

$$a_{100} = 2^{99} + 5 \times (2^{98} + 2^{97} + \dots + 1)$$

$$= 2^{99} + 5 \times 1 \times \frac{(2^{99} - 1)}{(2 - 1)} = 2^{99} + 5 \times 2^{99} - 5 = 6 \times 2^{99} - 5$$



16. C

$$\begin{aligned} & \frac{1}{(2^2 - 1)} + \frac{1}{(4^2 - 1)} + \frac{1}{(6^2 - 1)} + \dots + \frac{1}{(20^2 - 1)} \\ &= \frac{1}{(2+1)(2-1)} + \frac{1}{(4+1)(4-1)} + \dots + \frac{1}{(20+1)(20-1)} \\ &= 1/(1*3) + 1/(3*5) + 1/(5*7) + \dots + 1/(19*21) \\ &= 1/2 * (1/1 - 1/3 + 1/3 - 1/5 + 1/5 - 1/7 + \dots + 1/19 - 1/21) \\ &= 1/2 * (1 - 1/21) = 10/21 \end{aligned}$$

17. D

Let $x = 1$, so, $X_0 = 1$

$$X_1 = -1$$

$$X_2 = -1$$

$$X_3 = 1$$

$$X_4 = 1$$

$$X_5 = -1$$

$$X_6 = -1$$

So, X_n need not be positive when n is even, X_n need not be positive when n is odd, X_n need not be negative when n is even. So, none of the first three options are correct.

18. A

$$\text{Let } S = 2 + 5x + 9x^2 + \dots$$

$$S * x = 2x + 5x^2 + 9x^3 + \dots$$

$$S(1 - x) = 2 + 3x + 4x^2 + \dots$$

$$S(1 - x) * x = 2x + 3x^2 + 4x^3 + \dots$$

$$S(1 - x)(1 - x) = 2 + x + x^2 + x^3 + \dots = 2 + x/(1 - x)$$

$$\text{So, } S = [2(1 - x) + x]/(1 - x)^3 \Rightarrow S = (2 - x)/(1 - x)^3$$

19. D

If the child adds all the numbers from 1 to 34, the sum of the numbers would be $1+2+3+\dots+34 = 34*35/2 = 595$. Since the child got the sum as 575, he would have missed the number 20.

20. C

Given series can be written as:

$$\sum_{n=1}^{100} \left(\frac{1}{n \times (n+1)} \right)$$

$$\text{or } \sum_{n=1}^{100} \left(\frac{(n+1)-n}{n \times (n+1)} \right)$$

$$\text{or } \sum_{n=1}^{100} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

After putting values of n from 1 to 100, all terms will cancel out, only first and last terms will be there

$$\text{i.e. } 1 - \frac{1}{101}$$

$$\text{or } \frac{100}{101}$$

 VIDEO SOLUTION



21. A

$$\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$$

$$\text{or } \frac{2}{1-x^2} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$$

$$\text{or } \frac{4}{1-x^4} + \frac{4}{1+x^4}$$

$$\text{or } \frac{8}{1-x^8}$$

 VIDEO SOLUTION

22. A

$\frac{-p}{1-p}$ can be compared with sum of an infinite G.P. series i.e. $\frac{a}{1-r}$ (where a is first term and r is common ratio)

Hence here $a = (-p)$

and $r = p$

So k^{th} term will be $= (-p)(p)^{(k-1)}$

 VIEW SOLUTION

23. D

According to given question S_{50} will have 50 terms

And its first term will be 50th number in the series 1, 2, 4, 7, T_{50}

$$T_1 = 1$$

$$T_2 = 1 + 1$$

$$T_3 = 1 + 1 + 2$$

$$T_4 = 1 + 1 + 2 + 3$$

$$T_n = 1 + (1 + 2 + 3 + 4 + 5 \dots (n-1))$$

$$= 1 + \frac{n(n-1)}{2}$$

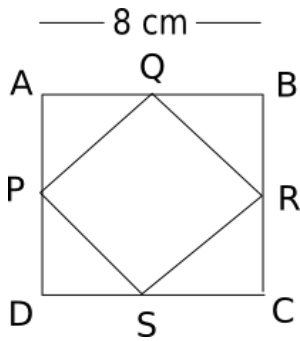
$$\text{So } T_{50} = 1 + 1225 = 1226$$

$$\text{Hence } S_{50} = (1226, 1227, 1228, 1229, \dots)$$

$$\text{And summation will be } = \frac{50}{2} (2 \times 1226 + 49 \times 1) = 62525$$

[VIDEO SOLUTION](#)

24. A



Side of first square = 8cm.

Side of second square made by joining mid-points of first square = $\frac{8}{\sqrt{2}}$

Similarly side of third square = $\frac{8}{\sqrt{2} \times \sqrt{2}}$ and so on.

Now summation of areas will be $8^2 + \left(\frac{8}{\sqrt{2}}\right)^2 + \left(\frac{8}{\sqrt{2} \times \sqrt{2}}\right)^2 + \dots$

or $8^2 \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right)$

or $64 \times \left(\frac{1}{1-\frac{1}{2}}\right)$ (As we know sum of an infinite G.P. is $\frac{a}{1-r}$ where a is first term and r is common ratio)

or 128 sq. cm.

[VIEW SOLUTION](#)

25. A

Suppose consecutive odd integers are: (a-2), a, (a+2)

Hence, $3a-6 = 2(a+2) + 3 \Rightarrow a=13$

$a+2 = 15$

[VIEW SOLUTION](#)



26. A

The seventh term of an AP = $a + 6d$. Third term will be $a + 2d$ and second term will be $a + d$. We are given that

$$(a + 6d)^2 = (a + 2d)(a + 16d)$$

$$\Rightarrow a^2 + 36d^2 + 12ad = a^2 + 18ad + 32d^2$$

$$\Rightarrow 4d^2 = 6ad$$

$$\Rightarrow d : a = 3 : 2$$

We have been asked about a:d. Hence, it would be 2:3

[VIDEO SOLUTION](#)

27. B

$a_1 = 3$ and $a_2 = 7$. Hence, the common difference of the AP is 4. If we assume, $k=3n$

We have been given that the sum up to $3n$ terms of this AP is 1830. Hence, $1830 = \frac{k}{2}[2 \times 3 + (k-1) \times 4]$

$$\Rightarrow 1830 \times 2 = k(6 + 4k - 4)$$

$$\Rightarrow 3660 = 2k + 4k^2$$

$$\Rightarrow 2k^2 + k - 1830 = 0$$

$$\Rightarrow (k-30)(2k+61) = 0$$

$$\Rightarrow k = 30 \text{ or } k = -61/2$$

Since k is the number of terms so k cannot be negative. Hence, must be 30

$$\text{So, } 3n = 30$$

$$n = 10$$

$$\text{Sum of the first '10' terms of the given AP} = 5 \times (6 + 9 \times 4) = 42 \times 5 = 210$$

$$m(a_1 + a_2 + \dots + a_n) > 1830$$

$$\Rightarrow 210m > 1830$$

$$\Rightarrow m > 8.71$$

Hence, smallest integral value of ' m ' is 9.

 VIDEO SOLUTION

28.51

$$\text{Sum of the sequence of even numbers is } 2a_3 + (2a_3 - 2) + (2a_3 - 4) + (2a_3 - 6) + (2a_3 - 8) = 450$$

$$\Rightarrow 10a_3 - 20 = 450$$

$$\Rightarrow a_3 = 47$$

$$\text{Hence } a_5 = 47 + 4 = 51$$

 VIDEO SOLUTION

29.C

Let the common ratio of the G.P. be r .

$$\text{Hence we have } a_n = 3(a_{n+1} + a_{n+2} + \dots)$$

The sum up to infinity of GP is given by $\frac{a}{1-r}$ where a here is a_{n+1}

$$\Rightarrow a_n = 3\left(\frac{a_{n+1}}{1-r}\right)$$

$$\Rightarrow a_n = 3\left(\frac{a_n \times r}{1-r}\right)$$

$$\Rightarrow r = \frac{1}{4}$$

$$\text{Now, } a_1 + a_2 + a_2 \dots = 32$$

$$\Rightarrow \frac{a_1}{1-r} = 32$$

$$\Rightarrow \frac{a_1}{3/4} = 32$$

$$\Rightarrow a_1 = 24$$

$$a_5 = a_1 \times r^4$$

$$a_5 = 24 \times (1/4)^4 = \frac{3}{32}$$

 VIDEO SOLUTION

30.A

$$a_{100} = \frac{1}{(3 \times 100 - 1) \times (3 \times 100 + 2)} = \frac{1}{299 \times 302}$$

$$\frac{1}{2 \times 5} = \frac{1}{3} \times \left(\frac{1}{2} - \frac{1}{5}\right)$$

$$\frac{1}{5 \times 8} = \frac{1}{3} \times \left(\frac{1}{5} - \frac{1}{8}\right)$$

$$\frac{1}{8 \times 11} = \frac{1}{3} \times \left(\frac{1}{8} - \frac{1}{11}\right)$$

....

$$\frac{1}{299 \times 302} = \frac{1}{3} \times \left(\frac{1}{299} - \frac{1}{302} \right)$$

$$\begin{aligned} \text{Hence } a_1 + a_2 + a_3 + \dots + a_{100} &= \frac{1}{3} \times \left(\frac{1}{2} - \frac{1}{5} \right) + \frac{1}{3} \times \left(\frac{1}{5} - \frac{1}{8} \right) + \frac{1}{3} \times \left(\frac{1}{8} - \frac{1}{11} \right) + \dots + \frac{1}{3} \times \left(\frac{1}{299} - \frac{1}{302} \right) \\ &= \frac{1}{3} \times \left(\frac{1}{2} - \frac{1}{302} \right) \\ &= \frac{25}{151} \end{aligned}$$

 VIDEO SOLUTION



31. C

Let $x = a$, $y = ar$ and $z = ar^2$

It is given that, $5x$, $16y$ and $12z$ are in AP.

so, $5x + 12z = 32y$

On replacing the values of x , y and z , we get

$$5a + 12ar^2 = 32ar$$

$$\text{or, } 12r^2 - 32r + 5 = 0$$

On solving, $r = \frac{5}{2}$ or $\frac{1}{6}$

For $r = \frac{1}{6}$, $x < y < z$ is not satisfied.

So, $r = \frac{5}{2}$

Hence, option C is the correct answer.

 VIDEO SOLUTION

32. C

Let 'x' be the average of all 52 positive integers $a_1, a_2, \dots, a_{52}$.

$$a_1 + a_2 + a_3 + \dots + a_{52} = 52x \dots (1)$$

Therefore, average of $a_2, a_3, \dots, a_{52} = x+1$

$$a_2 + a_3 + a_4 + \dots + a_{52} = 51(x+1) \dots (2)$$

From equation (1) and (2), we can say that

$$a_1 + 51(x+1) = 52x$$

$$a_1 = x - 51.$$

We have to find out the largest possible value of a_1 . a_1 will be maximum when 'x' is maximum.

$(x+1)$ is the average of terms $a_2, a_3, \dots, a_{52}$. We know that $a_2 < a_3 < \dots < a_{52}$ and $a_{52} = 100$.

Therefore, $(x+1)$ will be maximum when each term is maximum possible. If $a_{52} = 100$, then $a_{51} = 99, a_{50} = 98$ ends so on.

$$a_2 = 100 + (51-1)(-1) = 50.$$

$$\text{Hence, } a_2 + a_3 + a_4 + \dots + a_{52} = 50 + 51 + \dots + 99 + 100 = 51(x+1)$$

$$\Rightarrow \frac{51 * (50 + 100)}{2} = 51(x+1)$$

$$\Rightarrow x = 74$$

Therefore, the largest possible value of $a_1 = x - 51 = 74 - 51 = 23$.

VIDEO SOLUTION

33. 80707

$$S = 7 \times 11 + 11 \times 15 + 15 \times 19 + \dots + 95 \times 99$$

Nth term of the series can be written as $T_n = (4n + 3) * (4n + 7)$

Last term, $(4n+3) = 95$ i.e. $n = 23$

$$\sum_{n=1}^{n=23} (4n + 3) * (4n + 7)$$

$$\Rightarrow \sum_{n=1}^{n=23} 16n^2 + 40n + 21$$

$$\Rightarrow 16 * \frac{23 * 24 * 47}{6} + 40 * \frac{23 * 24}{2} + 21 * 23$$

$$\Rightarrow 80707$$

VIDEO SOLUTION

34. 105

Given that the arithmetic mean of x, y and z is 80.

$$\Rightarrow \frac{x + y + z}{3} = 80$$

$$\Rightarrow x + y + z = 240 \dots (1)$$

$$\text{Also, } \frac{x + y + z + v + u}{5} = 75$$

$$\Rightarrow \frac{x + y + z + v + u}{5} = 75$$

$$\Rightarrow x + y + z + v + u = 375$$

Substituting values from equation (1),

$$\Rightarrow v + u = 135$$

It is given that $u = (x+y)/2$ and $v = (y+z)/2$.

$$\Rightarrow (x + y)/2 + (y + z)/2 = 135$$

$$\Rightarrow x + 2y + z = 270$$

$$\Rightarrow y = 30 \text{ (Since } x + y + z = 240)$$

Therefore, we can say that $x + z = 240 - y = 210$. We are also given that $x \geq z$,

Hence, $x_{\min} = 210/2 = 105$.

VIDEO SOLUTION

35. 24

It is given that $t_1 + t_2 + \dots + t_n = 2n^2 + 9n + 13$, for every positive integer $n \geq 2$.

We can say that $t_1 + t_2 + \dots + t_k = 2k^2 + 9k + 13 \dots (1)$

Replacing k by $(k-1)$ we can say that

$$t_1 + t_2 + \dots + t_{k-1} = 2(k-1)^2 + 9(k-1) + 13 \dots (2)$$

On subtracting equation (2) from equation (1)

$$\Rightarrow t_k = 2k^2 + 9k + 13 - 2(k-1)^2 + 9(k-1) + 13$$

$$\Rightarrow 103 = 4k + 7$$

$$\Rightarrow k = 24$$

 VIDEO SOLUTION



36. **B**

$$a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n-1} a_n = n$$

It is clear from the above equation that when n is odd, the co-efficient of a is positive otherwise negative.

$$a_1 - a_2 = 2$$

$$a_1 = a_2 + 2$$

$$a_1 - a_2 + a_3 = 3$$

On substituting the value of a_1 in the above equation, we get

$$a_3 = 1$$

$$a_1 - a_2 + a_3 - a_4 = 4$$

On substituting the values of a_1, a_3 in the above equation, we get

$$a_4 = -1$$

$$a_1 - a_2 + a_3 - a_4 + a_5 = 5$$

On substituting the values of a_1, a_3, a_4 in the above equation, we get

$$a_5 = 1$$

So we can conclude that $a_3, a_5, a_7 \dots a_{n+1} = 1$ and $a_2, a_4, a_6 \dots a_{2n} = -1$

Now we have to find the value of $a_{51} + a_{52} + \dots + a_{1023}$

$$\text{Number of terms} = 1023 - 51 + 1$$

$$n = 973$$

There will be 486 even and 487 odd terms, so the value of $a_{51} + a_{52} + \dots + a_{1023} = 486 \cdot -1 + 487 \cdot 1 = 1$

 VIDEO SOLUTION

37. **4851**

Let us first find the number of terms

$$47 = 1 + (n-1)2$$

$$n = 24$$

$$24 \cdot 2n + 1 + 3 + 5 + \dots + 47 = 5280$$

$$48n + 576 = 5280$$

$$48n = 4704$$

$$n = 98$$

$$\text{Sum of first 98 terms} = 98 \cdot 99 / 2$$

$$= 4851$$

 VIDEO SOLUTION

38. B

A: 15, 19, 23, 27, , 415

B: 14, 19, 24, 29, . . . , 464

Here the first common term = 19

Common difference = LCM of 5, 4=20

$$19+(n-1)20 \leq 415$$

$$(n-1)20 \leq 396$$

$$(n-1) \leq 19.8$$

$$n=20$$

VIDEO SOLUTION

39. A

We have, $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_n}+\sqrt{a_{n+1}}}$

Now, $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} = \frac{\sqrt{a_2}-\sqrt{a_1}}{(\sqrt{a_2}+\sqrt{a_1})(\sqrt{a_2}-\sqrt{a_1})}$ (Multiplying numerator and denominator by $\sqrt{a_2} - \sqrt{a_1}$)

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1}$$

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{d} \quad (\text{where } d \text{ is the common difference})$$

Similarly, $\frac{1}{\sqrt{a_2}+\sqrt{a_3}} = \frac{\sqrt{a_3}-\sqrt{a_2}}{d}$ and so on.

Then the expression $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_n}+\sqrt{a_{n+1}}}$

can be written as $\frac{1}{d}(\sqrt{a_2} - \sqrt{a_1} + \sqrt{a_3} - \sqrt{a_2} + \dots + \sqrt{a_{n+1}} - \sqrt{a_n})$

$$= \frac{n}{nd}(\sqrt{a_{n+1}} - \sqrt{a_1}) \quad (\text{Multiplying both numerator and denominator by } n)$$

$$= \frac{n(\sqrt{a_{n+1}} - \sqrt{a_1})}{a_{n+1} - a_1} \quad (a_{n+1} - a_1 = nd)$$

$$= \frac{n}{\sqrt{a_1} + \sqrt{a_{n+1}}}$$

VIDEO SOLUTION

40. D

The population of town at the beginning of 1st year = p

The population of town at the beginning of 2nd year = 3+2p

The population of town at the beginning of 3rd year = 2(3+2p)+3 = 2*2p+2*3+3 = 4p+3(1+2)

The population of town at the beginning of 4th year = 2(2*2p+2*3+3)+3 = 8p+3(1+2+4)

Similarly population at the beginning of the nth year = $2^{n-1}p+3(2^{n-1} - 1) = 2^{n-1}(p + 3)-3$

The population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be $(2^{2034-2019})(1000 + 3)-3 = 2^{15}(1003)-3$

VIDEO SOLUTION



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41. 6144

11th term of series = a_{11} = Sum of 11 terms - Sum of 10 terms = $3(2^{11+1} - 2) - 3(2^{10+1} - 2)$
 $= 3(2^{12} - 2 - 2^{11} + 2) = 3(2^{11})(2 - 1) = 3 \cdot 2^{11} = 6144$

 VIDEO SOLUTION

42. A

$$x_1 = -1$$

$$x_1 = x_2 + 2 \Rightarrow x_2 = x_1 - 2 = -3$$

Similarly,

$$x_3 = x_1 - 5 = -6$$

$$x_4 = -10$$

.

.

The series is -1, -3, -6, -10, -15.....

When the differences are in AP, then the nth term is $-\frac{n(n+1)}{2}$

$$x_{100} = -\frac{100(100+1)}{2} = -5050$$

 VIDEO SOLUTION

43. D

Let the first term of the GP be "a". Now from the question we can show that

$$ar^{m-1} = \frac{3}{4} \quad ar^{n-1} = 12$$

Dividing both the equations we get $r^{m-1-n+1} = \frac{1}{16}$ or $r^{m-n} = 16^{-1}$ or $r^{n-m} = 16$

So for the minimum possible value we take Now give minimum possible value to "r" i.e -4 and n-m=2

Hence minimum possible value of r+n-m=-4+2=-2

 VIDEO SOLUTION

44. D

$$x_0 = 1$$

$$x_1 = 2$$

$$x_2 = \frac{(1+x_1)}{x_0} = \frac{(1+2)}{1} = 3$$

$$x_3 = \frac{(1+x_2)}{x_1} = \frac{(1+3)}{2} = 2$$

$$x_4 = \frac{(1+x_3)}{x_2} = \frac{(1+2)}{3} = 1$$

$$x_5 = \frac{(1+x_4)}{x_3} = \frac{(1+1)}{2} = 1$$

$$x_6 = \frac{(1+x_5)}{x_4} = \frac{(1+1)}{1} = 2$$

Hence, the series begins to repeat itself after every 5 terms. Terms whose number is of the form 5n are 1, 5n+1 are 2... and so on, where n=0,1,2,3,....

2021 is of the form 5n+1. Hence, its value will be 2.

 VIDEO SOLUTION

45. A

The first number in each group: 1, 2, 5, 10, 17.....

Their common difference is in Arithmetic Progression. Hence, the general term of the series can be expressed as a quadratic equation.

Let the quadratic equation of the general term be $ax^2 + bx + c$

1st term = $a+b+c=1$

2nd term = $4a+2b+c=2$

3rd term = $9a+3b+c=5$

Solving the equations, we get $a=1, b=-2, c=2$.

Hence, the first term in the 15th group will be $(15)^2 - 2(15) + 2 = 197$

We can see that the number of terms in each group is 1, 3, 5, 7.... and so on. These are of the form $2n-1$. Hence, the number of terms in 15th group will be 29. Hence, the last term in the 15th group will be $197+29-1 = 225$.

Sum of terms in group 15 = $\frac{29}{2} (197 + 225) = 6119$

Alternatively,

The final term in each group is the square of the group number.

In the first group 1, second group 4,

The final element of the 14th group is $(14)^2 = 196$, similarly for the 15th group this is : $(15)^2 = 225$

Each group contains all the consecutive elements in this range.

Hence the 15th group the elements are:

(197, 198,225).

This is an Arithmetic Progression with a common difference of 1 and the number of element 29.

Hence the sum is given by : $\frac{n}{2} \cdot (first\ term + last\ term)$

$\frac{29}{2} \cdot (197 + 225)$

6119.

 VIDEO SOLUTION



46. C

Given x, y, z are three terms in an arithmetic progression.

Considering $x = a, y = a+d, z = a+2*d$.

Using the given equation $x*y*z = 5*(x+y+z)$

$a*(a+d)*(a+2*d) = 5*(a+a+d+a+2*d)$

$= a*(a+d)*(a+2*d) = 5*(3*a+3*d) = 15*(a+d)$.

$= a*(a+2*d) = 15$.

Since all x, y, z are positive integers and $y-x > 2$, $a, a+d, a+2*d$ are integers.

The common difference is positive and greater than 2.

Among the different possibilities are : $(a=1, a+2d = 5), (a=3, a+2d = 5), (a=5, a+2d = 3), (a=15, a+2d = 1)$

Hence the only possible case satisfying the condition is :

$a = 1, a+2*d = 15$.

$$x = 1, z = 15.$$

$$z - x = 14.$$

 VIDEO SOLUTION

47. D

Now as per the given series :

$$\text{we get } x_1 = 1 + 2 = 3$$

$$\text{Now } x_1 - x_2 = 8$$

$$\text{so } x_2 = -5$$

$$\text{Now } x_1 - x_2 + x_3 = 15$$

$$\text{so } x_3 = 7$$

$$\text{so we get } x_n = (-1)^{n+1} (2n + 1)$$

$$\text{so } x_{49} = 99 \text{ and } x_{50} = -101$$

$$\text{Therefore } x_{49} + x_{50} = -2$$

 VIDEO SOLUTION

48. D

$$\text{Given } x_{n+1} = x_n + n - 1 \text{ and } x_1 = -1.$$

Considering

$$x_1 = -1. \quad (1)$$

$$x_2 = x_1 + 1 - 1 = x_1 + 0 \quad (2)$$

$$x_3 = x_2 + 2 - 1 = x_2 + 1 \quad (3)$$

$$x_4 = x_3 + 3 - 1 = x_3 + 2 \quad (4)$$

$$x_{100} = x_{99} + 98 \quad (100)$$

Adding the LHS and RHS for the hundred equations we have:

$$(x_1 + x_2 + \dots + x_{100}) = (-1 + 0 + \dots + 98) + (x_1 + x_2 + \dots + x_{99})$$

Subtracting this we have :

$$(x_1 + \dots + x_{100}) - (x_1 + \dots + x_{99}) = \frac{(98 \cdot 99)}{2} - 1.$$

$$x_{100} = 4851 - 1 = 4850$$

Alternatively

$$x_1 = -1$$

$$x_2 = x_1 + 1 - 1 = x_1 = -1$$

$$x_3 = x_2 + 2 - 1 = x_2 + 1 = -1 + 1 = 0$$

$$x_4 = x_3 + 3 - 1 = x_3 + 2 = 0 + 2 = 2$$

$$x_5 = x_4 + 4 - 1 = x_4 + 3 = 2 + 3 = 5$$

.....

If we observe the series, it is a series that has a difference between the consecutive terms in an AP.

Such series are represented as $t(n) = a + bn + cn^2$

We need to find $t(100)$.

$$t(1) = -1$$

$$a + b + c = -1$$

$$t(2) = -1$$

$$a + 2b + 4c = -1$$

$$t(3) = 0$$

$$a + 3b + 9c = 0$$

Solving we get,

$$b + 3c = 0$$

$$b + 5c = 1$$

$$c = 0.5$$

$$b = -1.5$$

$$a = 0$$

Now,

$$t(100) = (-1.5) 100 + (0.5) 100^2 = -150 + 5000 = 4850$$

 VIDEO SOLUTION

49. **C**

It is given,

$$S_n = 2n^2 + n$$

$$S_{n-1} = 2(n-1)^2 + (n-1)$$

$$S_{n-1} = 2n^2 - 3n + 1$$

$$T_n = S_n - S_{n-1} = 2n^2 + n - 2n^2 + 3n - 1 = 4n - 1$$

$$T_n = 4n - 1$$

The terms are 3, 7, 11, 15, 19, 23, 27,.....

27 is the first term in the series divisible by 9.

27 is the 7th term.

Therefore, the least possible value of n is 7.

The answer is option C.

 VIDEO SOLUTION

50. **14**

$$a_1 + a_2 + \dots + a_N = 300N$$

$$6a_1 + a_2 + \dots + a_N = 400N$$

$$5a_1 = 100N$$

$$a_1 = 20N$$

As the given sequence of numbers is non-decreasing sequence, N can take values from 2 to 15.

N is not equal to 1, if N = 1, then average of N numbers is 300 wouldn't satisfy.

Therefore, N can take values from 2 to 15, i.e. 14 values.

 VIDEO SOLUTION



51. A

Given, the number of particles on day 1 = 100

On day 2, one out of every 2 articles produces another particle.

The number of particles on day 2 will be $\frac{100}{2}$, i.e. 50 particles

On day 3, one out of every 3 articles produces another particle.

The number of particles on day 3 will be $\frac{100+50}{3}$, i.e. 50 particles

On day 4, one out of every 4 articles produces another particle.

The number of particles on day 4 will be $\frac{100+50+50}{4}$, i.e. 50 particles

Series will be 100, 50, 50, 50,....

It is given,

$$100 + (m-1)50 = 1000$$

$$m = 19$$

The answer is option A.

VIDEO SOLUTION

52. A

Sum of n terms in an A.P = $\frac{n}{2} (2a + (n-1)d)$

$$A_n = \frac{n}{2} (6 + (n-1)4) = n(2n+1)$$

$$\Sigma A_n = \Sigma n(2n+1) = 2\Sigma n^2 + \Sigma n = \frac{2n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

Substituting n = 25, we get

$$\frac{1}{25} \sum_{n=1}^{25} A_n = \frac{1}{25} \left(\frac{2(25)(25+1)(50+1)}{6} + \frac{25(25+1)}{2} \right)$$

$$\frac{1}{25} \sum_{n=1}^{25} A_n = 26(17) + 13 = 455$$

The answer is option A.

VIDEO SOLUTION

53. 548

$$\text{General term} = 38 + (n-1)17 = 17n + 21 = 17(n+1) + 4 = 17k + 4$$

Each term is in the form of $17k + 4$

Least 3-digit number in the form of $17k + 4$ is at $k = 6$, i.e. 106

Highest 3-digit number in the form of $17k + 4$ is at $k = 58$, i.e. 990

106, 123, 140,, 990

$$990 = 106 + 17(n-1)$$

$$n = 53$$

$$\text{Sum} = \frac{53}{2} (106 + 990) = 53 \times 548$$

$$\text{Average} = 53 \times \frac{548}{53} = 548$$

54. B

Given that $\frac{1}{\sqrt{y} + \sqrt{z}}$ is the arithmetic mean of $\frac{1}{\sqrt{x} + \sqrt{z}}$ and $\frac{1}{\sqrt{x} + \sqrt{y}}$

$$\Rightarrow \frac{2}{\sqrt{y} + \sqrt{z}} = \frac{1}{\sqrt{x} + \sqrt{z}} + \frac{1}{\sqrt{x} + \sqrt{y}}$$

$$\Rightarrow 2(\sqrt{x} + \sqrt{z})(\sqrt{x} + \sqrt{y}) = (\sqrt{y} + \sqrt{z})(\sqrt{x} + \sqrt{y} + \sqrt{x} + \sqrt{z})$$

$$\Rightarrow 2(x + \sqrt{xy} + \sqrt{xz} + \sqrt{yz}) = 2\sqrt{xy} + y + \sqrt{yz} + 2\sqrt{xz} + \sqrt{yz} + z$$

$$\Rightarrow 2x = y + z$$

$\Rightarrow y, x, z$ are in A.P. as x is the arithmetic mean of y and z .

55. 19

Given on day-1, there are 2 organisms.

On day-2, there are $2 \cdot 2 + 3 = 7$ and on day-3, there are $2 \cdot 7 + 3 = 17$.

Let us try to form a pattern:

$$2 = 2 + 0 \quad (n = 1)$$

$$7 = 4 + 3 \quad (n = 2)$$

$$17 = 8 + 9 \quad [8 + 3 \cdot 3] \quad (n = 3)$$

$$37 = 16 + 21 \quad [16 + 3 \cdot 7] \quad n = 4$$

$$T(n) = 2^n + 3(2^{n-1} - 1)$$

We know that $2^{20} = 2^{10} \times 2^{10} = 1024 \times 1024$, which is more than 1 million.

Let us check for $n = 19$

$$2^{19} + 3(2^{18} - 1) = 2^{19} + 3 \cdot 2^{18} - 3 = 2 \cdot 2^{19} + 2^{18} - 3 = 2^{20} + 2^{18} - 3, \text{ which is more than 1 million.}$$

Let us check for $n = 18$

$$\Rightarrow 2^{18} + 3(2^{17} - 1) = 2^{18} + 3 \cdot 2^{17} - 3 = 2 \cdot 2^{18} + 2^{17} - 3 = 2^{19} + 2^{17} - 3 \text{ which is not more than a million.}$$

$$\Rightarrow n = 19.$$

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56. D

It is given that $\frac{1}{2}$, $\frac{\log_3(2^x - 9)}{\log_3 4}$, and $\frac{\log_5(2^x + \frac{17}{2})}{\log_5 4}$ are in an arithmetic progression.

$$\frac{\log_3(2^x - 9)}{\log_3 4} \text{ can be written as } \log_4(2^x - 9), \text{ and } \frac{\log_5(2^x + \frac{17}{2})}{\log_5 4} \text{ can be written as } \log_4(2^x + \frac{17}{2})$$

$$\text{Hence, } 2 \log_4(2^x - 9) = \frac{1}{2} + \log_4(2^x + \frac{17}{2})$$

$$\frac{1}{2} \text{ can be written as } \log_4 2.$$

Therefore,

$$\Rightarrow 2 \log_4 (2^x - 9) = \log_4 2 + \log_4 \left(2^x + \frac{17}{2}\right)$$

$$\Rightarrow \log_4 (2^x - 9)^2 = \log_4 2 \left(2^x + \frac{17}{2}\right)$$

$$\Rightarrow (2^x - 9)^2 = 2 \left(2^x + \frac{17}{2}\right)$$

$$\Rightarrow 2^{2x} - 18 \cdot 2^x + 81 = 2 \cdot 2^x + 17$$

$$\Rightarrow 2^{2x} - 20 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^{2x} - 16 \cdot 2^x - 4 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^x (2^x - 16) - 4 (2^x - 16) = 0$$

$$\Rightarrow (2^x - 4) (2^x - 16) = 0$$

The values of 2^x can't be 4 (log will be undefined), which implies The value of 2^x is 16.

Therefore, the common difference is $\log_4 (2^x - 9) - \log_4 2$

$$\Rightarrow \log_4 7 - \log_4 2 = \log_4 \left(\frac{7}{2}\right)$$

The correct option is D

 VIDEO SOLUTION

57. D

The given sequence can be written as:

$$1 \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right) + \frac{1}{3} \left(\frac{1}{4} + \frac{1}{16} + \dots\right) + \frac{1}{9} \left(\frac{1}{16} + \frac{1}{64} + \dots\right) + \dots$$

We know that the sum of an infinite G.P. is $\frac{a}{1-r}$, where a is the first term and r is the common ratio.

$$\Rightarrow \text{The first term} = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$$

$$\Rightarrow \text{The second term} = \frac{1}{3} \left(\frac{\left(\frac{1}{4}\right)}{1-\left(\frac{1}{4}\right)}\right) = \frac{1}{9}$$

$$\Rightarrow \text{The third term} = \frac{1}{9} \left(\frac{\left(\frac{1}{16}\right)}{1-\left(\frac{1}{4}\right)}\right) = \frac{1}{108}$$

Observing these three terms, we see that they are in G.P. with a common ratio of $\frac{1}{12}$

$$\Rightarrow \text{Sum of this infinite G.P.} = \frac{\left(\frac{4}{3}\right)}{1 - \left(\frac{1}{12}\right)} = \frac{16}{11}$$

 VIDEO SOLUTION

58. A

The first series goes as follows:

46, 54, 62, 70, 78, 86, 94, 102,

The second series goes as follows:

98, 102, 106, 110, ...

The first common term is 102 (first term of the common terms) and the common difference between them will be $\text{hcf}(4, 8) = 8$

$\Rightarrow$ The required sequence is 102, 110, 118, (last term should be less than 468 (100th term of second series))

$$\Rightarrow 102 + (n-1)(8) \leq 498$$

$$\Rightarrow n \text{ is less than or equal to } 50.5 \Rightarrow n = 50$$

Using the summation of A.P. formula:

$$\text{Required sum} = \frac{n}{2} (2a + (n-1)d) = \frac{50}{2} (2 \times 102 + 49 \times 8) = 14900$$

 VIDEO SOLUTION

59. B

Let the first term of both series be a_1 , and b_1 , respectively, and the common difference be d_1 , and d_2 , respectively.

It is given that $a_5 = b_9$, which implies $a_1 + 4d_1 = b_1 + 8d_2$

$$\Rightarrow a_1 - b_1 = 8d_2 - 4d_1 \dots \text{Eq(1)}$$

Similarly, it is known that $a_{19} = b_{19}$, which implies $a_1 + 18d_1 = b_1 + 18d_2$

$$\Rightarrow a_1 - b_1 = 18d_2 - 18d_1 \dots \text{Eq(2)}$$

Equating (1) and (2), we get:

$$\Rightarrow 18d_2 - 18d_1 = 8d_2 - 4d_1$$

$$\Rightarrow 10d_2 = 14d_1$$

$$\Rightarrow 5d_2 = 7d_1$$

Since, d_1, d_2 are the prime numbers, which implies $d_1 = 5, d_2 = 7$.

It is also known that $b_2 = 0$, which implies $b_1 + d_2 = 0 \Rightarrow b_1 = -d_2 = -7$

Putting the value of b_1, d_1 , and, d_2 in Eq(1), we get:

$$a_1 = 8d_2 - 4d_1 + b_1 = 56 - 20 - 7 = 29$$

$$\text{Hence, } a_{11} = a_1 + 10d_1 = 29 + 10 \cdot 5 = 29 + 50 = 79$$

The correct option is B

 VIDEO SOLUTION

60. 967

It is given that $a_n = 13 + 6(n-1)$, which can be written as $a_n = 13 + 6n - 6 = 7 + 6n$

Similarly, $b_n = 15 + 7(n-1)$, which can be written as $b_n = 15 + 7n - 7 = 8 + 7n$

The common differences are 6, and 7, respectively, The common difference of terms that exists in both series is l.c.m (6, 7) = 42

The first common term of the first two series is 43 (by inspection)

Hence, we need to find the m th term, which is less than 1000, and the largest three-digit integer, and exists in both series.

$$t_m = a + (m-1)d < 1000$$

$$\Rightarrow 43 + (m-1)42 < 1000$$

$$\Rightarrow (m-1)42 < 957$$

$$\Rightarrow m-1 < 22.8 \Rightarrow m < 23.8 \Rightarrow m = 23$$

$$\text{Hence, the 23rd term is } 43 + 22 \times 42 = 967$$

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61. B

Opening the brackets, we get the series as: $\left(\frac{1}{5}\right)^2 - \left(\frac{1}{5} \times \frac{1}{7}\right) + \left(\frac{1}{5}\right)^4 - \left(\frac{1}{5} \times \frac{1}{7}\right)^2 + \left(\frac{1}{5}\right)^6 - \left(\frac{1}{5} \times \frac{1}{7}\right)^6 + \dots$

These are two infinite GPs when rearranged:

$$\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^4 + \left(\frac{1}{5}\right)^6 + \dots - \left(\frac{1}{5} \times \frac{1}{7}\right) - \left(\frac{1}{5} \times \frac{1}{7}\right)^6 - \left(\frac{1}{5} \times \frac{1}{7}\right)^2 - \dots$$

The sum of the first series would be $\frac{\frac{1}{25}}{1 - \frac{1}{25}} = \frac{1}{24}$

The sum of the second series would be $\frac{\frac{1}{35}}{1 - \frac{1}{35}} = \frac{1}{34}$

The answer to the given series would then be $\frac{1}{24} - \frac{1}{34} = \frac{10}{816} = \frac{5}{408}$

Therefore, Option B is the correct answer.

62. A

Using the arithmetic progression formula for the nth term, where

$$x_n = a + (n - 1)d$$

Substituting the value for n and using that in the equation that is given,

$$2x_6 + 2x_9 = x_{11} + x_{13}, \text{ Then, } x_{100} \text{ equals}$$

$$\text{We get, } 2(a + 5d) + 2(a + 8d) = a + 10d + a + 12d$$

$$4a + 26d = 2a + 22d$$

$$2a = -4d$$

$$a = -2d$$

$$\text{We are given, } x_5 = -4$$

$$a + 4d = -4$$

Substituting the value for a in terms of d,

$$2d = -4$$

$$d = -2$$

$$a = 4$$

$$x_{100} = a + 99d$$

$$x_{100} = 4 - 198 = -194$$

63.217

We are told that, for any natural number n_1 let a_n be the largest integer not exceeding $\sqrt{n}$

So for $n=1$, the largest integer not exceeding $\sqrt{1}$ will be 1

For $n=2$, the largest integer not exceeding $\sqrt{2}$ will be 1

For $n=3$, the largest integer not exceeding $\sqrt{3}$ will be 1

For $n=4$, the largest integer not exceeding $\sqrt{4}$ will be 2

We see a pattern here regarding the squares of the numbers,

Listing down all the perfect squares,

1, 4, 9, 16, 25, 36, 49, 64, ...

We see that the difference between 4 and 1 is 3 and there were three natural numbers in the given pattern with the value as 1,

So we can write for the rest of the numbers as well,

3 numbers will have value 1, giving a total value of 3

5 numbers will have value 2, giving a total value of 10

7 numbers will have value 3, giving a total value of 21

9 numbers will have value 4, giving a total value of 36

11 numbers will have value 5, giving a total value of 55

13 numbers will have value 6, giving a total value of 78

Now, only the values of a_{49} , a_{50} will have the value of 7, total value of 14.

Adding these values, we get the total sum as 217, which is the answer.

VIDEO SOLUTION

64.A

Finding the terms in the sequence, we see that $t_3 = 0, t_4 = -\frac{1}{3}, t_5 = 0$

We would notice that all the odd terms are 0, and we are also asked the sum of only even terms, so we do not need to consider those

$$t_6 = -\frac{1}{5}$$

We see that the even terms are in an HP: $-1, -\frac{1}{3}, -\frac{1}{5}, -\frac{1}{7}, \dots$

The sum we are asked is the inverse of these terms, that is: -1, -3, -5, -7, up to 1012 terms

The sum of this AP would be $\frac{[-(2 \times 1) + (1012 - 1)(-2)]}{2} \times 1012$

Which is equal to $-1012 \times 1012 = -1024144$

Therefore, Option A is the correct answer.

VIDEO SOLUTION

65.A

Given expression is $\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} < \{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$,

$$\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} = \{3^1 \times 3^4 \times 3^9 \dots \times 3^{400}\}$$

Sum of square of n natural numbers is $\frac{n \cdot (n+1) \cdot (2n+1)}{6}$

$$= 3^{\frac{(20 \cdot 21 \cdot 41)}{6}} = 3^{2870}$$

On right hand side of inequaltiy we have $\{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$

$$= 3^{21} \times 3^{22} \times \dots \times 3^{20+m} = 3^{21+22+\dots+20+m}$$

Using the sum of the first (n) natural numbers,

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$21 + 22 + \dots + (20 + m)$$

$$= 1 + 2 + \dots + (20 + m) - (1 + 2 + \dots + 20)$$

$$1 + 2 + \dots + (20 + m) = \frac{(20+m)(21+m)}{2}$$

$$1 + 2 + \dots + 20 = \frac{20 \cdot 21}{2} = 210$$

$$\text{So, } 21 + 22 + \dots + (20 + m)$$

$$= \frac{(20+m)(21+m)}{2} - 210$$

$$\text{Expanding, } (20 + m)(21 + m) = m^2 + 41m + 420$$

$$\text{Thus, } \frac{m^2 + 41m + 420}{2} - 210$$

$$= \frac{m^2 + 41m}{2}$$

Since the bases are equal, we must compare the powers.

$$2870 < \frac{m^2 + 41m}{2} \Rightarrow 5740 < m(m + 41)$$

Here, we can put in the option to check the minimum value that satisfies the inequality.

56: We get $5740 < 5264$. This is false

57: We get $5740 < 5586$. This is false

58: We get $5740 < 5742$. This is the minimum possible value.

 VIDEO SOLUTION

CAT Expected Questions PDF



66. A

The sum of all the elements in the given set = Sum of first 29 odd numbers = $29^2 = 841$

Let's assume that k is the m_{th} term. Sum of terms less than k = sum of first (m-1) odd numbers = $(m - 1)^2$

$$841 - m^2 = (m - 1)^2$$

$$841 - m^2 = m^2 - 2m + 1$$

$$840 - 2m^2 + 2m = 0$$

$$m^2 - m - 420 = 0$$

$$(m - 21)(m + 20) = 0$$

$$m = 21 - 20$$

$$m = 20. \text{ And the 20th term is } 2 \cdot m + 1 = 41$$

 VIDEO SOLUTION

67. 65

Let the first term of the arithmetic progression be a , and the common difference be d .

We have:

$$(a + 3d) + (a + 6d) + (a + 9d) = 99$$

$$\Rightarrow 3a + 18d = 99 \text{ or } a + 6d = 33 \dots (1)$$

We are also told that the sum of the first fourteen terms is 497,

$$\frac{14}{2}(2a + (14 - 1)d) = 497$$

$$\Rightarrow (2a + 13d) = \frac{497}{7}$$

$$\Rightarrow 2a + 13d = 71 \dots (2)$$

Solving equations (1) and (2), we get;

$$a = 3 \text{ and } d = 5$$

The first five terms would therefore be: 3, 8, 13, 18, 23, and their sum would be $3 + 8 + 13 + 18 + 23 = 65$.

[VIEW SOLUTION](#)

68.49

Let the copies sold on days 1 through 9 be $d_1, \dots, d_9$

From the averages:

$$d_1 + \dots + d_7 = 7 \times 60 = 420$$

$$d_1 + \dots + d_8 = 8 \times 63 = 504$$

$$\text{So } d_8 = 504 - 420 = 84. \text{ Then } d_9 = 84 - 11 = 73$$

The sum from day 2 to day 9 is $8 \times 66 = 528$

$$d_2 + \dots + d_9 = (d_1 + \dots + d_8) - d_1 + d_9 = 504 - d_1 + 73 = 577 - d_1$$

$$\text{Thus } 577 - d_1 = 528 \Rightarrow d_1 = 49$$

[VIDEO SOLUTION](#)

69.B

Let the first term be a and the common ratio be r .

Given

$$a + ar + ar^2 = 52 \quad \text{and} \quad a^2r + a^2r^2 + a^2r^3 = 624.$$

$$\text{From the first equation } a(1 + r + r^2) = 52, \text{ so } a = \frac{52}{1 + r + r^2}.$$

Substitute into the second:

$$\frac{52^2}{(1+r+r^2)^2} * (r + r^2 + r^3) = 624$$

$$\frac{r}{(1+r+r^2)} = \frac{3}{13}$$

$$3r^2 - 10r + 3 = 0$$

$$(3r - 1)(r - 3) = 0$$

Simplify to get an equation in (r) ; its real solutions are $r = \frac{1}{3}$ and $r = 3$. For an infinite geometric progression, we must have $(|r| < 1)$, so $r = \frac{1}{3}$.

Now

$$a = \frac{52}{1 + \frac{1}{3} + \frac{1}{9}} = \frac{52}{\frac{13}{9}} = 36$$

The sum of the infinite progression is

$$S = \frac{a}{1-r} = \frac{36}{1-\frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$$

 VIDEO SOLUTION

70. C

Assume the first series as $a, b, a/2, c, a+20$
and second series as x_1, x_2, x_3, x_4
 $x_1 = b - a$, $x_2 = a/2 - b$, $x_3 = c - a/2$, and $x_4 = a + 20 - c$
 $x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$
and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$
and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$
Solving we get, $a = 100$, $b = 60$, and $c = 70$
 $S_1 = 100, 60, 50, 70, 120$
 $S_2 = -40, -10, 20, 50$

 VIEW SOLUTION



71. A

Assume the first series as $a, b, a/2, c, a+20$
and second series as x_1, x_2, x_3, x_4
 $x_1 = b - a$, $x_2 = a/2 - b$, $x_3 = c - a/2$, and $x_4 = a + 20 - c$
 $x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$
and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$
and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$
Solving we get, $a = 100$, $b = 60$, and $c = 70$
 $S_1 = 100, 60, 50, 70, 120$
 $S_2 = -40, -10, 20, 50$

 VIEW SOLUTION

72. B

Assume the first series as $a, b, a/2, c, a+20$
and second series as x_1, x_2, x_3, x_4
 $x_1 = b - a$, $x_2 = a/2 - b$, $x_3 = c - a/2$, and $x_4 = a + 20 - c$
 $x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$
and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$
and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$
Solving we get, $a = 100$, $b = 60$, and $c = 70$
 $S_1 = 100, 60, 50, 70, 120$
 $S_2 = -40, -10, 20, 50$
Difference = 20

 VIEW SOLUTION

73. C

Assume the first series as $a, b, a/2, c, a+20$
and second series as x_1, x_2, x_3, x_4
 $x_1 = b - a$, $x_2 = a/2 - b$, $x_3 = c - a/2$, and $x_4 = a + 20 - c$
 $x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$
and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$

and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$
Solving we get, $a = 100$, $b = 60$, and $c = 70$
 $S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

Avg. $= (100 + 60 + 50 + 70 + 120) \div 5 = 80$

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74. **B**

Assume the first series as $a, b, a/2, c, a+20$
and second series as x_1, x_2, x_3, x_4
 $x_1 = b - a$, $x_2 = a/2 - b$, $x_3 = c - a/2$, and $x_4 = a + 20 - c$
 $x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$
and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$
and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$
Solving we get, $a = 100$, $b = 60$, and $c = 70$
 $S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

Sum = 20

[VIEW SOLUTION](#)

75. **D**

We will give an example to disprove each of the three options A, B and C and hence, the correct answer will be option D.

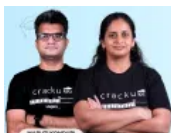
Initially, if the least integer in S_1 is -20 and the greatest integer in S_2 is 50, then after the doing the operations mentioned in the question, the greatest integer in S_1 is not greater than 50. Hence option A is false.

G is in S_2 as per the given information $\Rightarrow$ Option B is false.

If S_1 contains numbers from 1 to 24 and S_2 contains numbers from 25 to 50. Then $G = 50$ and $L = 1$. If all the numbers of S_1 change in sign, G will remain 50 and will be in S_2 while L will be -24 and will be in S_1 .

Hence, none of the statements is true always.

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76. **A**

We know that a_{24} is less than a_{25} .

So, even if a_{25} replaces a_{24} , the ascending order still exists in S_1 .

But, a_{25} is less than a_{26} . Hence, the descending order does not exist in S_2 anymore.

[VIEW SOLUTION](#)

77. **D**

For the least element L in S_1 to be greater than the greatest element or equal to G in S_2 , the number that is added to L cannot be less than $G - L$.

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